



School of Computer Science and Engineering

Faculty of Engineering

The University of New South Wales

SlicerBoneMat: A 3D Slicer Extension for CT-Based Material Mapping

by

Maxwell Hogan

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Supervisor: Dr. Luca Modenese

Student ID: z5417656

Abstract

Finite element analysis of bone, using material properties derived from computed tomography data, is a well-established approach for patient-specific biomechanical modelling, with applications in fracture risk assessment, implant design, and surgical planning. The predominant tool for this material mapping step (assigning spatially varying Young's modulus values to finite element meshes from CT intensity data) is Bonemat, a closed-source, Windows-only executable with no programming interface and limited recent development. At present, equivalent functionality is not available within 3D Slicer, the most widely used open-source medical imaging platform.

This thesis presents SlicerBoneMat, an open-source Python extension for 3D Slicer that implements automated CT-based material mapping to volumetric finite element meshes of bone. The extension supports configurable density-elasticity relationships, multiple algorithm variants, and direct export to FEBio, Abaqus, Ansys, and VTK formats.

Validation against Bonemat V3.2 on a proximal femur specimen yielded an RMSE of 0.4212 MPa (less than 0.01% of the mean assigned modulus) with a correlation coefficient and regression slope both effectively equal to 1.0. A sensitivity analysis showed that varying the integration step count introduces modulus variability approximately 152 times larger than the tool-to-tool difference, a relationship that, to the author's knowledge, has not been previously quantified for this class of software.

SlicerBoneMat demonstrates close agreement with Bonemat while addressing several of its core limitations. Through its direct FEBio export, it supports a complete, free, open-source, cross-platform pipeline from CT image to biomechanical simulation within a single workflow. SlicerBoneMat is publicly available through the 3D Slicer Extensions Manager.

Contents

1	Introduction	1
2	Background and Related Work	5
2.1	Clinical Significance of CT-based Bone Modelling	5
2.2	The Material Mapping Pipeline	6
2.3	Bonemat	7
2.4	MITK-GEM	8
2.5	Proprietary Material Mapping Solutions	9
2.6	FE Solvers and Open-Source Alternatives	10
2.7	FEBio Studio	10
2.8	3D Slicer as the Integration Platform	11
3	Methods/Implementation	14
3.1	System Overview and Architecture	14
3.2	User Interface and Configurable Parameters	15
3.2.1	Inputs	15
3.2.2	The Density Conversion Chain	16
3.2.3	Phantom Calibration Table	17
3.2.4	Algorithm Selection	17
3.2.5	Additional Parameters	17

3.2.6	Mesh Statistics Panel	18
3.2.7	Output and Export	19
3.3	Input Handling and Validation	19
3.3.1	Mesh Validation	19
3.3.2	CT Coordinate System Handling	20
3.4	The Material Mapping Algorithm	21
3.4.1	Algorithm V1: HU Node Averaging	22
3.4.2	Algorithm V2: HU Field Integration	22
3.4.3	Algorithm V3: Young’s Modulus Field Integration	22
3.4.4	Numerical Integration Scheme	23
3.5	Modulus Binning	24
3.6	Export Pipeline	24
3.6.1	VTK	25
3.6.2	Abaqus (.inp)	25
3.6.3	Ansys (.cdb)	25
3.6.4	FEBio (.feb)	25
3.7	Visualisation	26
3.8	Summary of Design Decisions	27
4	Results	29
4.1	Experimental Setup	29
4.1.1	Bone Specimens	29
4.1.2	CT Data	30
4.1.3	Density-Elasticity Parameters	31
4.1.4	Software and Hardware Environment	33
4.1.5	Test Case Overview	33

4.2	Material Assignment Accuracy – Proximal Femur (T1, T2)	34
4.2.1	Descriptive Statistics	34
4.2.2	Equivalence with Bonemat (T1)	35
4.2.3	Integration Step Sensitivity (T1 vs T2)	37
4.3	Material Assignment Accuracy – Full Femur (T3)	41
4.3.1	Descriptive Statistics	41
4.3.2	Equivalence with Bonemat	42
4.4	Binning Accuracy (T4)	46
4.4.1	Binning Statistics	47
4.4.2	Bin Assignment Residuals	48
4.4.3	Effect of Binning on the Modulus Distribution	49
4.5	Computational Performance (T5)	50
4.5.1	Computation Time vs Mesh Size (T5a)	50
4.5.2	Computation Time vs Integration Steps (T5b)	51
4.6	Mesh Statistics Accuracy (T6)	52
4.6.1	Summary Statistics	53
4.6.2	Distribution Comparison	54
4.7	Summary of Validation Results	56
5	Discussion	57
5.1	Equivalence with Bonemat	57
5.2	The Influence of <i>numSteps</i>	59
5.3	Positioning Against Existing Tools	60
5.4	Limitations	62
5.5	Future Work	63
6	Conclusions	64

List of Figures

4.1	Scatterplot of T1	36
4.2	Residual scatterplot of T1	37
4.3	Scatterplot of T1 against T2	39
4.4	Residual scatterplot of T1 against T2	40
4.5	Overlapping histogram of T1 and T2	41
4.6	Scatterplot of T3	44
4.7	Residual scatterplot of T3	45
4.8	Qualitative comparison of material assignment on full femur	46
4.9	Residual scatterplot of T4	48
4.10	Comparison of modulus distribution for T1 against T4	49
4.11	Scatterplot of T5a	51
4.12	Scatterplot of T5b	52
4.13	Mesh statistic distribution grid for SBM against FEBio Studio	55

List of Tables

4.1	Bone specimen information	30
4.2	CT data information	30
4.3	Constant experiment parameters	32
4.4	Test case overview	33
4.5	Descriptive statistics for T1 and T2	34
4.6	Comparative statistics for T1	35
4.7	Comparative statistics for T1 against T2	37
4.8	Descriptive statistics for T3	42
4.9	Comparative statistics for T3	42
4.10	Binning statistics	47
4.11	Mesh statistics for SBM vs FEBio Studio	53
4.12	Mesh statistics for SBM vs FEBio Studio	56

Chapter 1

Introduction

Bone is one of the most mechanically important tissues in the human body. As the structural scaffold supporting all human movement, bone is simultaneously a living, adaptive material and a frequent site of clinical failure from factors like fracture, degradation, and pathological change. Understanding the mechanical behaviour of bone is therefore not just an academic exercise; it has direct implications for fracture risk assessment, the design and validation of orthopaedic implants, pre-surgical planning, and the biomechanical modelling of populations at elevated injury risk, including the elderly, athletes, and soldiers [ZMV99].

The primary computational method used to characterise bone mechanics is finite element analysis (FEA). Since its first application to bone biomechanics in 1972 [BPS72], FEA has evolved into the standard tool for simulating stress and strain distributions within skeletal structures. The process begins with a computed tomography (CT) scan of the target bone. The CT image encodes spatially varying Hounsfield unit (HU) values, which can be related, through established relationships, to Young's modulus, which is the fundamental measure of a material's stiffness [CH77, HPS⁺08]. By discretising the bone geometry into a volumetric finite element mesh and assigning these CT-derived modulus values to each element, a patient-specific model is produced that reflects both the geometry and the mechanical heterogeneity of that individual's bone.

Applied in this way, FEA can predict fracture location and failure load [KRJS97], evaluate the biomechanical consequences of implant placement, and simulate surgical scenarios before they are performed in the clinic.

The accuracy and utility of such models is directly dependent on the quality of the material mapping step (the process of converting CT intensity data into spatially-distributed mechanical properties assigned to the mesh). Several software tools have been developed to perform this operation. Bonemat, first released in 1998 by researchers at the Istituto Ortopedico Rizzoli in Bologna, Italy [ZMV99], is the one of the most widely used research tool for this purpose, and remains the de facto standard against which other implementations are compared. However, Bonemat is a standalone executable built on a legacy C++ framework. It exposes no programming API, receives limited updates, is not open-source, and can be difficult to integrate into modern medical imaging workflows.

On the commercial side, FE solvers such as Ansys and Abaqus are the most widely used environments for bone stress analysis, yet neither provides native CT-based material mapping; instead, researchers must rely on external preprocessing tools to prepare their models. Platforms such as Materialise Mimics [Mat26] (bundled as part of the Mimics Innovation Suite) and Synopsys Simpleware ScanIP [Syn26] offer that material mapping functionality alongside comprehensive segmentation and meshing pipelines. These tools are capable and well supported, but carry substantial licence costs, with figures in the range of several thousand euros per year reported in the literature, which places them out of reach for many research groups and institutions in low-resource settings [Res13]. Both commercial platforms are also Windows-centric, limiting their accessibility to users on macOS or Linux systems.

MITK-GEM [PFB⁺16] represents a more recent open-source alternative, but has several limitations. It provides a complete open-source pipeline from image segmentation through tetrahedral mesh generation to density-to-modulus material mapping, and its configurable power-law interface makes it the most functionally comparable free alternative to Bonemat. However, its build system has strict dependency constraints,

requiring Qt 5.5 specifically, with Qt 5.6 explicitly unsupported, and additionally suffers from limited documentation, narrow file format support, and poor integration with the broader medical imaging ecosystem. Additionally, MITK-GEM’s documentation states that it is for research use only and has not been validated against in-vitro strength measurements or cohort databases, which further limits its viability.

One platform well positioned to address these deficiencies is 3D Slicer [KPV14]. Originally developed at the Surgical Planning Laboratory at Brigham and Women’s Hospital and the MIT Artificial Intelligence Laboratory, 3D Slicer is a free, open-source software platform for medical image visualisation, segmentation, and analysis. By the time of its 5.0 release in 2022, 3D Slicer had accumulated over one million downloads of the core program and 5.3 million downloads of its extensions over the preceding decade. It runs on Windows, macOS, and Linux, and is built on foundational scientific libraries including VTK and ITK. Extensions can be developed in Python, enabling rapid prototyping and broad accessibility to the research community. Despite this, no extension currently exists within the 3D Slicer ecosystem to perform automated CT-based material mapping to bone finite element meshes.

This thesis aims to address that gap. The aim of this project is to develop SlicerBoneMat, an open-source extension for 3D Slicer that implements the material mapping capabilities of Bonemat within the 3D Slicer environment. The extension enables users to (1) import a volumetric mesh derived from CT segmentation (performed upstream within the 3D Slicer ecosystem) or from a finite element package, (2) perform automated material mapping of Young’s modulus from CT data to the mesh using established density–elasticity relationships, and (3) export the mapped mesh for use in downstream finite element solvers. The implementation is validated against Bonemat outputs using two human femur specimens, with additional sensitivity analysis on the effect of the Gauss integration step count on mapping accuracy and computational cost.

The extension supports tetrahedral (linear and quadratic), linear wedge, and linear hexahedral element types, with per-element assignment of isotropic Young’s modulus derived from clinical CT data. However, validation in this work is conducted on tetrahe-

dral meshes of femur specimens only; generalisation to other element types and anatomical sites is left for future work. Anisotropic material models, quantitative CT (QCT) calibration, and cortical-specific mapping (as implemented in Bonemat V4 [SPFT20]) are acknowledged as potential extensions to the current implementation but fall outside the scope of this work.

The remainder of this report is structured as follows. Chapter 2 provides background on the medical imaging pipeline and the theoretical basis for CT-derived material mapping, alongside a review of the existing literature and software landscape. Chapter 3 describes the design and implementation of SlicerBoneMat. Chapter 4 presents the validation results and sensitivity analysis. Chapter 5 discusses the findings in the context of existing tools and identifies directions for future work. Chapter 6 draws conclusions.

Chapter 2

Background and Related Work

2.1 Clinical Significance of CT-based Bone Modelling

Bone fractures are a major global health burden, with osteoporotic hip fractures alone carrying a one-year mortality rate approaching 20–30% in older adults [AN15], and incidence projected to increase as populations age [Wor23]. Patient-specific finite element (FE) models derived from computed tomography (CT) data have become a widely used non-invasive tool for fracture risk assessment. CT-based FE models have been shown to outperform dual-energy X-ray absorptiometry (DXA) (the current clinical standard) at predicting bone strength *ex vivo*, and perform at least equivalently at identifying individuals at risk of vertebral fracture in clinical cohort data [IE22]. Continued refinements to meshing strategies, anatomical referencing, and loading configurations have further increased predictive accuracy [MMPX18]. Beyond fracture risk, these models support implant design, pre-surgical planning, and biomechanical research. Despite this, model generation remains largely confined to specialist groups, partly due to the fragmentation of the required software pipeline.

2.2 The Material Mapping Pipeline

Producing a patient-specific FE bone model from CT data involves five main steps.

HU acquisition. CT scanners reconstruct the X-ray attenuation at each spatial location throughout the scanned volume into a three-dimensional grid of voxels. These attenuation coefficients are expressed in Hounsfield units (HU), a linearly scaled measure defined such that water = 0 HU and air = -1000 HU [Hou73]. Local HU values reflect tissue density and form the basis for material property estimation.

Phantom calibration. HU values are scanner-dependent due to beam hardening and acquisition differences [DIE⁺24]. A linear calibration $\rho_{CT} = \alpha \cdot \text{HU} + \beta$, fitted against a phantom of known hydroxyapatite concentration, converts HU to bone mineral density (ρ_{CT}) [LKJ19]. SlicerBoneMat accepts two user-supplied HU-to-density reference points and fits this relationship internally, delegating calibration to the user’s existing workflow.

Ash density and apparent density. As demonstrated by Schileo et al. [SDT⁺08], ρ_{CT} is first converted to ash density (ρ_{ash}), the density of the mineral content remaining after the organic matrix is removed by incineration, using the equation $\rho_{ash} = (\rho_{CT} + 0.09) / 1.14$. The conversion reflects the fact that CT attenuation specifies mineral content specifically, not total tissue mass. Ash density is then related to apparent density (ρ_{app}), the bulk density of the bone inclusive of pore space, through the empirical relationship $\rho_{app} = \rho_{ash} / 0.6$. Schileo et al. highlighted that this two-stage conversion, rather than a direct HU-to-apparent-density shortcut, produces more accurate subject-specific FE model predictions.

Young’s modulus. Apparent density is related to Young’s modulus E through power-law relationships of the form $E = k\rho^n$, first formalised by Carter and Hayes [CH77]. However, the experimentally-derived constants k and n vary by anatomical site, bone type, and experimental protocol. A review of 18 studies [HPS⁺08] found 22 distinct elasticity–density relationships with no clear consensus, discussing how even after normalising for methodological differences, significant inter-study discrepancies remained,

with no universally accepted relationship. Morgan et al. [MBK03] further showed that site-specificity is a significant source of inter-study variation, with relationships calibrated for the femur performing poorly when applied to the spine and vice versa. Both Bonemat and SlicerBoneMat expose this choice to the user rather than hardcoding a single formulation.

Mesh assignment. The spatially-varying Young’s modulus field is assigned to mesh elements via Gauss quadrature sampling of the CT-derived modulus at integration points within each element. Chen et al. [CWL⁺15] extensively compared element-based and nodal-based strategies, and produced equivalent global mechanical results, with the main differences being in programmatic difficulty.

2.3 Bonemat

BoneMat was developed in 1998 by researchers at the Istituto Ortopedico Rizzoli in Bologna, Italy, as the first purpose-built tool for automated CT-based material property assignment to finite element bone meshes [ZMV99]. It accepts a CT image and a volumetric tetrahedral or hexahedral mesh, performs the HU-to-modulus conversion chain described in Section 2.2, and outputs a mesh with per-element Young’s modulus values in formats ready for FE analysis. Three major algorithmic versions have been published, each refining the core mapping strategy.

In v1, the CT lattice is overlaid on the mesh, and the Young’s modulus is calculated independently at each CT node falling inside an element, with the element’s final modulus obtained by averaging those values. For elements containing no CT nodes, modulus is estimated from the eight CT vertices surrounding the element centroid.

In v2, rather than averaging moduli, a single representative density for each element is first obtained by numerical integration of the density field across the element volume, and Young’s modulus is then calculated from that single density value [TPV04]. The approach uses all available CT information and is more accurate, at the cost of additional computation.

In v3, the density field is first transformed into a continuous Young’s modulus field, and integration is then performed over that modulus field rather than the density field [THP⁺08]. This subtle but important change (integrating in modulus space rather than density space) was shown to significantly reduce average and peak strain prediction errors while maintaining strong correlation with experimental measurements [TSH⁺07]. Subsequent minor releases (v3.1 in 2015, v3.2 in 2016) added a graphical interface, an Ansys importer/exporter, and an Abaqus interface via a collaboration with Pegg et al. [PG16].

Despite its foundational importance, Bonemat has several limitations that restrict its use in modern research workflows. The software has not received a major algorithmic update since 2007. It is closed-source, offering no avenue for researchers with specialised requirements to extend or inspect its implementation. It runs exclusively on Windows, excluding researchers on macOS or Linux systems. Its supported output formats (Ansys and Abaqus) are themselves commercially licensed, meaning that using Bonemat end-to-end requires access to proprietary solvers. There is no active developer community or public issue tracker, and the user manual, absent until v3.1, remains sparse. These barriers collectively restrict Bonemat’s integration into modern, reproducible research pipelines.

2.4 MITK-GEM

MITK-GEM [PFB⁺16] is a free, open-source pipeline built atop the Medical Imaging Interaction Toolkit (MITK), providing a complete workflow from CT segmentation through volumetric meshing to material property assignment. Its primary innovation at release was a graph-cut segmentation algorithm that reduced femur segmentation time from approximately 20 minutes of manual effort to 2–5 minutes of interactive work, while producing FE results statistically comparable to manual segmentation. Its material mapping plugin converts Hounsfield unit values through a configurable power-law pipeline to produce a volumetric mesh with elastic modulus assigned to each node and element, saved as an unstructured grid file.

However, MITK-GEM has an important limitation: the software explicitly states it lacks validation against in-vitro strength measurements and cohort databases, and its results cannot be used for diagnosing, preventing, or treating medical conditions. For a tool with the potential to replace Bonemat in research contexts, the absence of published quantitative validation of the mapping algorithm against experimental ground truth is a clear gap.

Beyond this, MITK-GEM presents considerable practical barriers. Its build system imposes strict dependency constraints (Qt 5.5 is required, with Qt 5.6 explicitly unsupported), making compilation from source difficult on modern systems. Its native output is a VTK unstructured grid (.vtu) file, and conversion to FE solver formats such as Abaqus or Ansys requires separately maintained MATLAB scripts that are provided as examples but are not integrated into the tool. Finally, MITK-GEM is built for the MITK Workbench, a developer-oriented toolkit rather than an application-focused platform, which makes it poorly suited to researchers seeking integration with broader medical imaging workflows.

2.5 Proprietary Material Mapping Solutions

Several commercial platforms offer material mapping as part of a broader medical imaging suite. Materialise Mimics, part of the Mimics Innovation Suite [Mat26], and Synopsys Simpleware ScanIP [Syn26] both provide CT segmentation, mesh generation, and heterogeneous material assignment within polished, well-supported graphical environments. These tools are technically capable and actively maintained, but carry annual licensing costs in the range of several thousand euros, placing them out of reach for many research groups and institutions in lower-resource settings. Both platforms are also predominantly Windows-based, and their closed-source nature prevents methodological transparency or researcher-led extension.

In-house implementations represent a third category. Research groups have published custom scripted approaches to material assignment, typically MATLAB or Python

scripts that operate on CT data and mesh files to replicate Bonemat’s core algorithm for a specific solver or workflow [FZK21]. While such approaches give researchers full methodological control, they are rarely generalised, documented, or distributed, meaning each group bears the development cost independently and reproducibility across studies is limited.

2.6 FE Solvers and Open-Source Alternatives

The downstream consumers of material-mapped bone meshes are FE solvers, the software that actually performs the stress and strain calculations. The main commercial options are Ansys [ANS26] and Abaqus [Das26], both of which offer mature, general-purpose nonlinear FE capabilities with large user bases in engineering and biomechanics. Bonemat v3.2 supports export directly to both formats. However, both require expensive commercial licences, and neither provides native CT-to-modulus material mapping as they are solvers, not imaging pipelines. The material assignment must be performed upstream, making tools like Bonemat a prerequisite for their use in bone FE modelling.

This dependency on commercial infrastructure, both for the solver and, until recently, for the material mapping tool, creates a notable access barrier for the broader biomechanics research community. The response has been FEBio.

2.7 FEBio Studio

FEBio (Finite Elements for Biomechanics) [MEAW12] is a free, open-source nonlinear FE solver developed jointly by the University of Utah and Columbia University, specifically designed for biomechanical applications. It supports large deformation quasi-static and dynamic structural mechanics, multiphasic materials, fluid mechanics, and fluid-solid interaction, with constitutive models tailored to biological tissue behaviour [MW25]. It is distributed under the MIT licence and runs on Windows, ma-

cOS, and Linux. FEBio Studio is the accompanying graphical application, providing a GUI for constructing FEBio input files, running simulations, and visualising results. It has become the primary open-source FE environment for biomechanics research.

FEBio Studio does provide some material mapping functionality via an "image map" data generator, which allows users to associate CT image intensity with element properties. However, this functionality is rudimentary compared to Bonemat: it does not implement the multi-stage $HU \rightarrow BMD \rightarrow \text{ash density} \rightarrow \text{apparent density} \rightarrow \text{Young's modulus}$ conversion chain, nor does it offer configurable density-elasticity relationships or Gauss integration-based element sampling. Users wishing to apply heterogeneous CT-derived properties must either manually edit the XML-based *.feb* input file, or use workarounds involving image map data generators that do not replicate the rigour of established material mapping algorithms. Furthermore, FEBio Studio does not support CT image segmentation, meaning users must bring a pre-segmented mesh to the tool. Its strict XML format also creates friction when importing meshes generated in other environments.

The practical consequence is that FEBio, despite being the preferred open-source solver in biomechanics, lacks a companion tool for the upstream material mapping step, a gap that motivates dedicated preprocessing software capable of producing FEBio-compatible output.

2.8 3D Slicer as the Integration Platform

3D Slicer (hereafter Slicer) is a free, open-source platform for the visualisation, processing, segmentation, registration, and analysis of medical images and meshes, distributed under a BSD-style licence with no restrictions on use [FBKC⁺12, 3D 26]. By the time of its 5.0 release in 2022, Slicer had accumulated over one million downloads of the core program and 5.3 million downloads of its extensions over the preceding decade. It supports Windows, macOS, and Linux, and handles all major medical imaging formats including DICOM. The platform has been under continuous development since 1999,

with an active international developer and user community, regular minor releases, and responsive community forums.

Slicer’s architecture follows a modular and layered approach. The core application provides fundamental infrastructure (data management, rendering, I/O, and a Python console) atop foundational libraries including VTK (visualisation), ITK (image processing), and Qt (GUI). Functionality is delivered through modules: self-contained units each implementing a specific task, selectable from the main interface. Core modules covering segmentation, registration, and quantitative analysis are distributed with the application; additional extensions are available through Slicer’s Extension Manager, an indexed catalogue of community-developed modules that have passed maintainability and documentation standards.

Extensions can be implemented in Python or C++; the 5.0 release improved integration with the Python scientific ecosystem, allowing direct installation of packages from PyPI within Slicer’s embedded Python environment [HSFR22]. This makes Slicer a particularly accessible development platform for researchers comfortable with Python (the same language used extensively in scientific computing and data analysis) without requiring knowledge of the C++ core. The combination of VTK, ITK, NumPy, and PyPI access within a single environment means that image processing, mesh manipulation, and numerical computation can all be performed without leaving the platform.

Importantly, Slicer already supports the full imaging pipeline from CT import through segmentation and mesh generation, with the latter performed via extensions such as SlicerSegmentMesher. A missing step in its ecosystem is the automated assignment of CT-derived material properties to those meshes. Existing Slicer community forum discussions document demand for this functionality from researchers attempting to replicate Bonemat workflows within Slicer, but no published extension addresses it. SlicerBoneMat is designed to address this gap: implemented as a Python extension within Slicer’s established architecture, it brings the full Bonemat material mapping pipeline into the platform where users already perform their segmentation and meshing, eliminating the need to export data to a separate tool and then reimport results.

Combined with FEBio Studio for downstream simulation, this creates a complete, free, open-source, cross-platform bone FE modelling workflow, from CT scan to stress analysis, without dependence on any commercial software licence.

Chapter 3

Methods/Implementation

3.1 System Overview and Architecture

SlicerBoneMat is implemented as a Python-scripted extension of 3D Slicer, following the *ScriptedLoadableModule* architecture provided by the Slicer framework. The extension is composed of three distinct layers, each with a clearly defined responsibility.

The outermost layer is the Slicer module infrastructure, comprising the *BoneMat* class (module registration), *BoneMatParameterNode* (parameter persistence), and *BoneMatWidget* (graphical interface). These classes handle module registration within the Slicer environment, rendering of the Qt-based user interface, connection of UI signals to handler functions, and persistence of user-defined parameters across sessions using Slicer’s parameter node mechanism. This layer contains no computation; its sole responsibility is mediating between the user and the computation layer beneath it.

The middle layer is *BoneMatLogic*, a class that inherits from *ScriptedLoadableModuleLogic* and implements all computation independently of the graphical interface. This separation allows the logic to be used independently of the interface, meaning it can be imported and invoked from a Python script or Jupyter notebook without instantiating any graphical components. As such, batch processing and programmatic integration is

enabled for developers, which is not possible in Bonemat.

The innermost layer is a vendored, modified copy of *py_bonemat_abaqus*, the open-source Python reimplementation of the Bonemat algorithm developed by Pegg and Gill [PG16] for Abaqus users. Rather than depending on an externally installed package, which would create version management problems in Slicer’s embedded Python environment, the library is included as a subtree within the extension repository and loaded explicitly at import time. A dedicated function, *configurePyBonematImports()*, ensures that the vendored copy takes priority in Python’s module search path over any system-installed version, and invalidates the module cache to prevent stale imports across reloads. This ensures consistent behaviour across environments, regardless of local Python configuration.

The overall data flow through the system is illustrated conceptually as follows: the user supplies a CT volume and a volumetric mesh within Slicer, configures the conversion parameters, and initiates the mapping. *BoneMatLogic.process()* extracts a coordinate-corrected CT scalar field and a typed mesh representation, delegates per-element modulus computation to the mapping engine, bins the resulting moduli, attaches them to the mesh as cell data, and renders the result in the Slicer viewport. The mapped mesh can then be exported in the user’s chosen format.

3.2 User Interface and Configurable Parameters

The user interface is defined in a Qt Designer *.ui* file and loaded at module initialisation. It is organised into logical sections, each controlling a distinct aspect of the mapping process.

3.2.1 Inputs

The user selects a CT volume and a volumetric mesh from dropdown selectors populated with nodes already present in the Slicer scene. The CT volume must be a

vtkMRMLScalarVolumeNode, which is the standard Slicer representation for 3D scalar imaging data. The mesh must be a *vtkMRMLModelNode* wrapping a *vtkUnstructuredGrid*, which is the VTK data structure for volumetric finite element meshes. Both of these are standard Slicer node types, meaning that CT data loaded via the DICOM importer and meshes imported via any of Slicer’s mesh import extensions (or meshed within Slicer using the *SlicerSegmentMesher* extension) are immediately compatible inputs without format conversion.

3.2.2 The Density Conversion Chain

The core of the parameter interface exposes the full five-step conversion chain described in Section 2.2. The parameters are:

- **CT density slope and intercept:** the linear coefficients of the calibration line $\rho_{CT} = \alpha \cdot \text{HU} + \beta$, converting raw Hounsfield unit values to bone mineral density in g/cm^3 .
- **Ash density offset and scale:** the linear parameters of the CT-to-ash-density relationship, following the formulation of Schileo et al. [SDT⁺08]
- **Apparent density divisor:** a scaling factor applied during the ash-to-apparent density conversion. These three parameters together define the complete lambda function passed to the mapping engine.
- **Modulus scale and exponent:** the constants k and n in the power-law relationship $E = k\rho^n$, relating apparent density to Young’s modulus in the style of Carter and Hayes [CH77].

These seven parameters are exposed as spin boxes with fine-grained step control. A preset system allows users to save and reload named configurations, persisted to Qt application settings and available across Slicer sessions. Several presets based on published density-elasticity relationships are included by default, covering common formulations of the power law regression between density and modulus from the literature [MBK03].

SlicerBoneMat allows researchers to switch between validated parameter sets with a single dropdown selection. In comparison, Bonemat requires either manual editing of the parameters each run, or externally saving a parameters file of unknown format.

3.2.3 Phantom Calibration Table

An optional phantom calibration interface allows users to derive the CT density slope and intercept directly from physical measurements, rather than supplying them as abstract numerical values. When enabled, the user enters two pairs of known HU-to-density reference values (obtained from scanning a calibration phantom) into a two-row table. The extension computes the linear regression slope and intercept from these two points and automatically populates the CT density spin boxes. This reflects the workflow described in Section 2.2: the calibration is performed externally by the researcher, and SlicerBoneMat consumes its output. Enabling the phantom table locks the slope and intercept spin boxes to prevent inconsistency.

3.2.4 Algorithm Selection

A dropdown allows the user to select from the three Bonemat algorithm variants (V1, V2, V3), defaulting to V3. The algorithmic differences between these variants are described fully in Section 3.4.

3.2.5 Additional Parameters

- **Number of integration steps:** controls the density of the quadrature grid used in V2 and V3 integration. Higher values increase accuracy at the cost of runtime. This parameter is the subject of the sensitivity analysis in Chapter 4.
- **Poisson's ratio:** a constant applied uniformly to all elements in the exported mesh. While SlicerBoneMat maps heterogeneous Young's modulus, Poisson's

ratio is treated as homogeneous, consistent with standard practice in bone FE modelling.

- **Minimum modulus:** a floor value applied after computation, preventing any element from being assigned a near-zero or physically unrealistic modulus.
- **Gap value:** controls the width of modulus bins during post-processing grouping, described in Section 3.5.
- **CT padding:** an optional voxel buffer of user-specified HU value added around the boundary of the CT volume, described in Section 3.3.

3.2.6 Mesh Statistics Panel

An interactive mesh statistics panel visualises the distribution of element volumes, tetrahedral quality scores, and minimum and maximum edge lengths as histograms, computed on demand when the user loads a mesh. This allows the user to assess mesh quality before initiating the potentially time-consuming mapping step. FEBio Studio provides equivalent mesh quality metrics, and a direct comparison of these statistics between the two tools is presented in Chapter 4.

Note that tetrahedral mesh quality is quantified using the radius-edge ratio (the ratio of the circumsphere radius to the shortest edge length of each element). The theoretical minimum of approximately 0.612 corresponds to a perfect equilateral tetrahedron [Si13], and larger values indicate increasingly degenerate elements. To ensure direct comparability with FEBio Studio in the validation study of Chapter 4, the radius-edge ratio was implemented by transcribing FEBio Studio’s C++ source calculation into Python. No equivalent Python implementation was publicly available, enabling direct comparison with FEBio Studio metrics.

3.2.7 Output and Export

The user selects an output format from a dropdown: VTK (*.vtk*), FEBio (*.feb*), Abaqus (*.inp*), or Ansys (*.cdb*). Following successful mapping, the Download button opens a file dialog and serialises the result in the chosen format. The export implementations are described in Section 3.6.

3.3 Input Handling and Validation

Before any mapping computation is performed, *BoneMatLogic.process()* applies a sequence of validation checks that produce informative error messages on failure.

3.3.1 Mesh Validation

The input mesh is first checked to confirm it is a volumetric mesh rather than a surface mesh, a common mistake given that surface meshes are the more typical output of segmentation workflows. The mesh is then checked for homogeneity: all elements must be of the same type, as mixed-element meshes are not supported, consistent with Bonemat’s behaviour. Finally, the element type is confirmed to be one of the four supported types: linear tetrahedral (4 nodes), quadratic tetrahedral (10 nodes), linear wedge (6 nodes), or linear hexahedral (8 nodes). Supporting these 4 element geometries makes SlicerBoneMat compatible with a wider range of mesh generation tools and FE solver preferences.

Prior to mapping, the bounding box of all mesh nodes is compared against the extent of the CT volume. If any node falls outside the CT data range, an error is raised identifying the specific node index and coordinate. This check is essential: the integration algorithm will produce undefined results for elements that sample outside the CT grid, and a clear diagnostic message is considerably more useful than a silent numerical failure.

3.3.2 CT Coordinate System Handling

The CT coordinate handling is a central piece of original engineering in the extension, and warrants detailed description.

Slicer stores all imaging data in RAS (Right-Anterior-Superior) physical space, using an IJK-to-RAS affine matrix to describe the relationship between the voxel grid indices and physical coordinates. The mapping engine, inherited from *py.bonemat.abaqus*, expects a rectilinear coordinate grid – three ascending arrays of physical coordinates (x, y, z) and a flat array of scalar values indexed consistently with those coordinates. Constructing this representation correctly from a Slicer volume node requires careful handling of the affine transformation, coordinate ordering, and potential axis orientations.

The *getVtkCTData()* function performs this construction in several steps:

- **Axis alignment check.** The IJK-to-RAS matrix is first inspected to confirm the CT volume is axis-aligned, done by enforcing that each IJK axis maps predominantly to a single RAS axis with no substantial off-diagonal rotation. If the CT is obliquely oriented (as can occur with certain scanner protocols or after registration), the extension raises an error directing the user to resample the volume to an axis-aligned orientation using Slicer’s existing resampling tools before proceeding. Since oblique CT resampling introduces its own interpolation errors and edge cases, implementing it internally would have added considerable complexity without clear benefit.
- **Coordinate array construction.** The IJK-to-RAS matrix is applied to the voxel index grid to produce physical coordinate arrays in RAS space. Separate arrays are constructed for each axis by applying the matrix to unit steps along that axis.
- **Axis direction normalisation.** The code detects axes where the coordinate array runs in descending order and reverses both the coordinate array and the corresponding dimension of the scalar volume. This ensures the rectilinear grid

always presents ascending coordinate arrays, as required by the bisection-based lookup in *vtk_data.find_grid()*. Failing to handle this case would produce silently incorrect spatial lookups for a significant proportion of real clinical CT datasets.

- **CT boundary padding.** An optional user-configurable padding layer is applied around all six faces of the CT volume. When enabled, additional voxel slices of a user-specified HU value (typically -1000, representing air) are prepended and appended along each axis. This solves the practical edge case of a mesh node falling precisely on or fractionally outside the CT boundary (due to floating point rounding during mesh generation), at which point the bounding box check would fail and prevent mapping. Padding extends the CT domain by a small margin, allowing these boundary elements to be mapped without error. The padding value is user-configurable because appropriate values differ depending on the segmentation context: a femur embedded in surrounding tissue should be padded with tissue-density HU values, not air.

The resulting coordinate arrays and flattened, reordered scalar data are packaged into a *vtk_data* object from the mapping engine, which provides the spatial lookup and interpolation methods used during integration.

3.4 The Material Mapping Algorithm

The material mapping is performed by *_assign_mat_props()* in the vendored *calc.py*, invoked by *BoneMatLogic.process()*. Three algorithm variants are implemented, corresponding to Bonemat V1, V2, and V3 respectively. The user selects the algorithm variant and the 7 configurable density-to-modulus parameters at run time. The *get_Params()* method in *BoneMatLogic* translates those parameters into the format consumed by the mapping engine, combining the ash density offset, scale, and apparent density divisor into the linear coefficient pair expected by the conversion functions, and applying a floor of *minVal* at each stage to prevent negative intermediate values from causing errors in the power-law step.

3.4.1 Algorithm V1: HU Node Averaging

In V1, the CT lattice is conceptually overlaid on the mesh. For each element, `vtk_data.get_voxels()` identifies all CT lattice nodes whose coordinates fall within the element’s bounding box and pass an interior-point test. The HU values at those nodes are averaged, and the mean HU is passed through the conversion chain once to yield a single modulus value for the element.

For elements smaller than a single CT voxel (a common occurrence when mesh resolution exceeds CT resolution) the centroid of the element is located, and the eight surrounding CT nodes are used instead. This is the behaviour inherited from the original Bonemat V1 formulation. V1 is the computationally cheapest variant and is restricted to linear tetrahedral elements, as the interior-point test is only implemented for that geometry.

3.4.2 Algorithm V2: HU Field Integration

V2 replaces the node-averaging approach with numerical integration over the element volume. The integration sweeps through the element in natural coordinates, sampling the CT scalar field at each quadrature point via trilinear interpolation, accumulating a volume-weighted sum of HU values. The result is a single representative HU for the element, which is then passed through the full conversion chain once.

This approach uses all available CT information within the element volume rather than only the subset of CT nodes that happen to fall within it, and produces more accurate density estimates particularly when mesh elements span multiple CT voxels.

3.4.3 Algorithm V3: Young’s Modulus Field Integration

V3 is the most rigorous variant and the default selection. It differs from V2 mathematically, as rather than integrating the HU field and applying the conversion chain once

at the end, the full conversion chain is applied at every quadrature point before integration. The element’s modulus is then the volume-averaged result of those per-point modulus values.

This distinction matters because the density-to-modulus conversion is nonlinear, with the power-law relationship meaning that the modulus of the mean density is not equal to the mean of the moduli. V3 correctly computes the latter, which Taddei et al. [THP⁺08] demonstrated produces significantly lower average and peak strain prediction errors compared to V2.

3.4.4 Numerical Integration Scheme

The integration in V2 and V3 steps through each element’s internal volume using three nested loops across the element’s parametric coordinate space, sampling the CT scalar field at regular intervals. At each sampling point, a geometric correction factor (the Jacobian determinant) is applied to account for the distortion between the element’s idealised parametric shape and its actual physical geometry in space. The accumulated samples are then volume-normalised to yield a single representative value for the element. This scheme is inherited directly from Bonemat and *py-bonemat_abaqus* and is preserved unchanged to maintain numerical equivalence with the reference implementation.

For a tetrahedral element, the number of sampling points is proportional to n^3 where n is the value of *numSteps*, since the three nested loops each scale with n . Therefore, doubling *numSteps* increases the number of samples per element by a factor of approximately eight, and computation time scales accordingly. The *numSteps* parameter’s effect on mapping accuracy and runtime is characterised in the validation study of Chapter 4.

3.5 Modulus Binning

Following per-element modulus computation, `_limit_num_materials()` applies an optional post-processing step that groups nearby modulus values into shared bins. This step is controlled by the `gapValue` parameter.

The binning algorithm sorts all computed moduli in ascending order, then works from the largest value downward, assigning each element the maximum modulus value within a window of width `gapValue`. Elements whose moduli fall within `gapValue` MPa of each other are assigned a common value, reducing the total count of distinct moduli in the output. The implementation of this step in SlicerBoneMat doesn't offer users the choice between max or mean grouping density, instead enforcing the former, more conservative choice from a structural safety standpoint, avoiding underestimation of local stiffness.

FE solvers define material properties as a list of named sections, each with a distinct modulus value, and thus a large bone mesh without binning may produce thousands of unique modulus values, each requiring a separate material definition in the output file. Abaqus in particular can experience excessive preprocessing overhead with very large numbers of material sections. In practice, binning reduces this to a manageable number without meaningfully affecting the spatial variation of the material field, provided `gapValue` is chosen to be small relative to the range of moduli in the model. Setting `gapValue = 0` disables binning entirely, preserving the full resolution of the mapping.

3.6 Export Pipeline

The mapped moduli are stored as an array attached to the cell data of the output `vtkUnstructuredGrid`. This is Slicer's standard mechanism for associating scalar fields with mesh elements, and makes the modulus data immediately available for visualisation using Slicer's scalar display infrastructure. The export step serialises this data to disk in the user's chosen format.

3.6.1 VTK

The VTK format export uses VTK’s built-in *vtkUnstructuredGridWriter*. This produces a file readable by VTK-based analysis tools, other Slicer extensions and most image-based platforms. It is the most information-preserving format and is recommended for users who wish to inspect or further process the mapped mesh before committing to a solver-specific format.

3.6.2 Abaqus (*.inp*)

The Abaqus exporter writes a valid Abaqus input file structured for immediate use in a static analysis. Elements sharing the same binned modulus value are grouped into named element sets, each assigned a corresponding material definition with the appropriate Young’s modulus and Poisson’s ratio. This provides the Abaqus export functionality previously available only through Bonemat V3.2 or the standalone *py_bonemat_abaqus* package, now integrated directly into the Slicer workflow.

3.6.3 Ansys (*.cdb*)

The Ansys exporter produces a command file readable by Ansys Mechanical, defining node coordinates, element connectivities, and material properties. Elements are grouped by their assigned modulus value, with each group written as a separate material with its own Young’s modulus and Poisson’s ratio. The appropriate Ansys element type is selected automatically based on the mesh’s element geometry.

3.6.4 FEBio (*.feb*)

The FEBio exporter produces a valid FEBio 4.0 XML specification file, implementing a single isotropic-elastic material with Young’s modulus assigned per element via FEBio’s mapped parameter mechanism. Elements are grouped by modulus into named element

sets, and each set’s modulus values are listed in a corresponding data block referencing a shared *moduli_map* identifier. The structural mechanics module (solid) is declared, and all node coordinates and element connectivities are written.

As noted in Section 2.7, FEBio Studio does not provide a streamlined path for assigning CT-derived heterogeneous material properties: users must either manually edit the *.feb* XML file or use an undocumented image map workflow. The FEBio exporter in SlicerBoneMat produces a *.feb* file that FEBio Studio can open and execute directly, completing the free, open-source pipeline from CT scan through material mapping to biomechanical simulation without any manual file editing or format conversion. Furthermore, using the mapped parameter mechanism with a single material, rather than individual materials for each binned modulus value, allows the material distribution to be visualised using the Mesh Inspector tool, which gives users additional visualisation and statistical options within FEBio Studio.

3.7 Visualisation

Upon successful completion of the mapping step, the output mesh is automatically rendered in the Slicer 3D viewport with per-element scalar colouring by Young’s modulus value. A red-to-blue colour map is applied, with scalar range set automatically to the data extent. A colour legend is added to the viewport with a label of “Young’s Modulus” and formatted tick labels showing modulus values to the nearest integer.

This immediate visual feedback serves two purposes. First, it allows the user to verify that the mapping has produced a spatially plausible result, with both the characteristic gradient from low-modulus trabecular bone at the interior and higher-modulus cortical bone at the surface being visually apparent. Second, it supports quality assurance: unexpected spatial artefacts such as sharp discontinuities at the CT boundary, or a uniform single colour indicating a failed calibration producing near-identical moduli throughout, are immediately detectable without exporting and loading the file in a separate application.

3.8 Summary of Design Decisions

Several architectural decisions are outlined below.

Python over C++. Slicer extensions can be implemented in either Python or C++. Python was chosen primarily for accessibility, as researchers in the biomechanics community are far more likely to be comfortable reading and modifying Python than C++, making the open-source contribution more reusable and the build and testing pipeline much simpler by skipping complex compilation. NumPy provides sufficient numerical performance for the mesh sizes encountered in clinical bone modelling. C++ would offer faster execution for the integration loop, but the bottleneck in practice is the per-element CT sampling, which is fundamentally serial and limited by the quadrature step count rather than language overhead.

Vendor rather than depending on an external package. Including a modified copy of *py-bonemat_abaqus* directly within the repository, rather than depending on it as an installed package, made it possible to make targeted changes to the library's internals that substantially simplified the integration with Slicer. The original *py-bonemat_abaqus* was designed as a file-based pipeline: CT and mesh data were read from DICOM/VTK and Abaqus input files, respectively, into internal data structures, processed, and then written back out to new files. Integrating this as-is into Slicer would require serialising Slicer's in-memory mesh representation to a temporary file on disk before reloading it, which is an unnecessary round-trip that adds complexity and I/O overhead. By vendoring the library, the file import and export routines could be bypassed entirely, with mesh data passed directly from Slicer's VTK data structures into the mapping engine's internal classes. The binning implementation was also replaced with a more efficient algorithm. These changes would not have been possible without direct access to the library's source, and maintaining them as patches against an external dependency would have been fragile and difficult to manage across package updates.

Axis-alignment requirement for CT input. Oblique CT volumes could in principle

be supported by applying the full affine transformation at each spatial lookup during integration. However, the rectilinear grid assumption in `vtk_data.find_grid()`, which uses bisection search along independent x, y, z arrays, is fundamental to the performance of the lookup. Replacing it with a fully general affine lookup would require restructuring the mapping engine substantially and would introduce new interpolation accuracy considerations. Delegating oblique CT resampling to Slicer’s existing, validated resampling infrastructure is both simpler and more robust.

Preservation of the uniform quadrature scheme. The uniform mid-point quadrature inherited from `py_bonemat_abaqus` is preserved unchanged in order to maintain numerical equivalence with Bonemat, which is the reference implementation against which this project is validated. Substituting a different quadrature scheme, even a theoretically superior one, would make it impossible to distinguish algorithmic differences from implementation differences in the validation results.

Chapter 4

Results

The implementation described in Chapter 3 is evaluated through a structured validation study addressing four questions: whether SlicerBoneMat produces material assignments equivalent to Bonemat under matched conditions; whether the integration step parameter meaningfully affects that agreement; whether the computational cost of the extension scales predictably with mesh size and integration density; and whether the mesh quality statistics reported by the extension agree with those produced by FEBio Studio.

4.1 Experimental Setup

4.1.1 Bone Specimens

Two bone specimens are used across the validation tests (T1-T4). The first is a proximal femur, used as the primary equivalence test case and for all sensitivity and computation time experiments. Its smaller size makes it practical for repeated runs across a range of parameters, and it is the most common use-case for the material-mapping workflow. The second is a full femur, used to test whether the equivalence result obtained on the proximal specimen generalises to a geometrically more complex and larger mesh. Basic

mesh properties for each specimen are summarised in Table 4.1.

Table 4.1: Bone specimen information

Properties	Proximal Femur	Full Femur
Number of nodes	264354	375696
Number of elements	187258	261699
Element type	Linear tetrahedron	Linear tetrahedron
Approx.bounding box dimensions (mm)	$114.7 \times 90.9 \times 146.7$	$105.3 \times 148.8 \times 460.4$

4.1.2 CT Data

The two specimens were derived from CT scans of different patients, sourced independently. The proximal femur CT was obtained from the MITK-GEM public dataset, available in its tutorial, as an isotropic resampled volume with a voxel spacing of $1 \times 1 \times 1$ mm and no accompanying acquisition metadata [Ara17]. The full femur CT was sourced from the publicly available HFValid dataset [ALMS⁺25] from the Istituto Ortopedico Rizzoli in Italy, the same institution who created the Bonemat software. Acquisition metadata for both specimens is summarised in Table 4.2.

Table 4.2: CT data information

Properties	Proximal Femur	Full Femur
Scanner manufacturer	–	GE Medical Systems
Scanner model	–	HISpeed CT/I
Pixel spacing	1 mm	0.7422 mm
Spacing between slices	1 mm	2 mm
Slice thickness	1 mm	3 mm
KVP	–	120 kV
X-ray tube current	–	200 mA
Calibration coefficient (intercept)	0.041036	-0.004792
Calibration coefficient (slope)	0.000586	0.000806

In both cases the CT volume was loaded into 3D Slicer as a scalar volume node via the standard DICOM importer, and the same file was subsequently provided as input to Bonemat V3.2, ensuring both tools operated on identical imaging data for each respective test case. The use of two independent CT datasets introduces a confounding

variable when comparing equivalence results across specimens, and thus any difference in agreement between the proximal and full femur tests may reflect differences in CT acquisition protocol as well as differences in bone geometry. This is discussed further in Section 5.

4.1.3 Density-Elasticity Parameters

The parameters governing the material mapping pipeline are separated into two categories: CT calibration coefficients, which convert raw Hounsfield unit values to bone mineral density and are specimen-specific, and density-elasticity parameters, which define the subsequent conversion from density to Young’s modulus and are held constant across all test cases.

CT calibration coefficients. For the full femur specimen, the phantom calibration coefficients provided directly in the HFValid dataset metadata were used, derived from asynchronous calibration against a European Spine Phantom at the Rizzoli Orthopaedic Institute. These are the same coefficients used by the HFValid authors when generating their own Bonemat-mapped FE models from this dataset, providing direct traceability between the calibration inputs used here and those of the originating study.

For the proximal femur specimen, no phantom calibration data was available, as the MITK-GEM dataset provides the CT volume as an isotropic resampled image without accompanying acquisition metadata or phantom measurements. Calibration coefficients were instead derived from two known tissue reference points identified directly within the scan. The maximum HU value observed within the scan was identified and attributed to cortical bone, which has a known mineral density of 1.2 g/cm^3 of hydroxyapatite equivalent. The minimum HU value observed within the femoral shaft was identified and attributed to bone marrow, which has an effective mineral density of approximately 0 g/cm^3 . A linear calibration line was fitted through these two reference points to yield the coefficients used for this specimen. This approach is a form of internal tissue-based calibration, an established alternative to phantom-based calibration when phantom data is unavailable, though it introduces greater uncertainty than a rig-

ously controlled phantom measurement, and this is acknowledged as a limitation of the proximal femur test case. The resulting calibration coefficients for both specimens are reported in Table 4.2.

Shared density-elasticity parameters. The ash density, apparent density, modulus scale and exponent, Poisson’s ratio, minimum modulus, and integration algorithm were held constant across all material assignment test cases.

The ash and apparent density conversion parameters were taken from Schileo et al. [SDT⁺08], whose formulation was demonstrated to improve the accuracy of subject-specific FE models over direct HU-to-apparent-density approaches. The power-law modulus coefficients were taken from Morgan et al. [MBK03], which provides site-specific relationships for several anatomical locations; the femoral neck relationship was applied to both specimens as the closest reported anatomical site to the regions of interest. Whilst the choice of site-specific relationship has a meaningful effect on the absolute modulus values assigned, it does not affect the comparative equivalence between SlicerBoneMat and Bonemat, as both tools were run with identical coefficients. The validation is therefore insensitive to this choice.

Identical parameter values were entered into both SlicerBoneMat and Bonemat for each test case, ensuring that any differences in output between the two tools are attributable to algorithmic implementation rather than parameterisation. The parameter values are outlined in Table 4.3.

Table 4.3: Constant experiment parameters

Constant parameters	Value
Ash density offset	0.09
Ash density scale	1.14
Apparent density divisor	0.6
Modulus scale	6850
Modulus exponent	1.49
Algorithm	Modulus integration (v3)
Minimum modulus (MPa)	10
Poisson’s ratio	0.35

4.1.4 Software and Hardware Environment

Bonemat V3.2 was run on Windows 11, emulated on macOS Sequoia via Parallels Desktop 18. SlicerBoneMat was run within 3D Slicer version 5.10 natively on the same macOS machine. All computation time measurements were recorded on a machine with a 10-core M2 Pro chip with 16GB of RAM, with no other computationally intensive processes running concurrently. Timing within 3D Slicer was measured as wall-clock elapsed time from the moment the Apply button was pressed to the moment the mapping completed, using Python’s *time* module. Bonemat’s runtime was recorded manually using a digital stopwatch; as Bonemat completed all test cases in under 5 seconds (considerably faster than SlicerBoneMat) precise timing was not considered necessary for the purposes of comparison, and runtime figures for Bonemat are reported as approximate values.

4.1.5 Test Case Overview

Eight test cases are conducted across the validation study, summarised in Table 4.4. Tests T1 through T4 assess material assignment accuracy against Bonemat under varying conditions. Tests T5a and T5b characterise computational performance. Test T6 evaluates mesh statistics accuracy against FEBio Studio.

Table 4.4: Test case overview

Test	Specimen	<i>numSteps</i>	<i>gapValue</i>	Purpose
1	Proximal femur	3	1	Primary equivalence
2	Proximal femur	4	1	Integration step sensitivity
3	Full femur	3	1	Secondary equivalence
4	Proximal femur	3	50	Binning analysis
5a	Varies	3	1	Computation time vs. mesh size
5b	Proximal femur	Varies	1	Computation time vs. <i>numSteps</i>
6	Proximal femur	—	—	Mesh statistics accuracy

For all accuracy tests, per-element Young’s modulus values from SlicerBoneMat and Bonemat are compared using mean absolute error (MAE), root mean square error

(RMSE), Pearson correlation coefficient R^2 , and the slope of a linear regression of SlicerBoneMat moduli against Bonemat moduli. A slope close to 1.0 and R^2 close to 1.0 together indicate strong element-wise agreement. RMSE is weighted more heavily than MAE in the interpretation of results, as it penalises large localised disagreements more severely, a property that matters for FE accuracy where a small number of outlier elements with incorrect moduli can meaningfully affect predicted stress concentrations.

Note that for brevity, SlicerBoneMat and Bonemat are abbreviated to SBM and BM respectively in all figure labels and table headers throughout this chapter.

4.2 Material Assignment Accuracy – Proximal Femur (T1, T2)

Two test cases are conducted on the proximal femur specimen: T1, comparing SlicerBoneMat against Bonemat at $numSteps = 3$, and T2, running SlicerBoneMat at $numSteps = 4$ to assess the sensitivity of the material assignment to the integration step parameter. Bonemat is not run for T2, as T1 already establishes the equivalence between the two tools; T2 is therefore a within-SlicerBoneMat comparison against T1.

4.2.1 Descriptive Statistics

Table 4.5 presents the descriptive statistics of the assigned Young’s modulus distributions across all elements for Bonemat, SlicerBoneMat T1, and SlicerBoneMat T2.

Table 4.5: Descriptive statistics for T1 and T2

Statistic (MPa)	Bonemat	SlicerBoneMat (T1)	SlicerBoneMat (T2)
Mean	3600.9	3600.9	3589.2
Min	334.5	334.0	334.0
Max	16201.2	16202.0	15933.0
Range	15866.7	15868.0	15599.0
Median	2320.9	2321.0	2323.0

The distributions produced by Bonemat and SlicerBoneMat T1 are in close agreement across all reported statistics. The mean modulus differs by less than 0.1 MPa, the minimum values are within 0.5 MPa of each other, and the medians are virtually identical at 2320.9 MPa and 2321.0 MPa respectively. This similarity across the full distributional range (from the minimum trabecular values through to the cortical maximum) suggests that SlicerBoneMat reproduces not just the central tendency of Bonemat’s output but the shape of the entire material distribution.

SlicerBoneMat T2 shows slightly more divergence from T1. The mean decreases from 3600.9 MPa to 3589.2 MPa, and the maximum drops from 16202.0 MPa to 15933.0 MPa, a reduction of approximately 269 MPa at the upper end of the distribution. The minimum and median remain essentially unchanged. This pattern suggests that increasing the integration step count from 3 to 4 primarily affects the highest-modulus elements, which correspond to cortical bone regions where HU gradients are steepest and integration accuracy is most sensitive to sampling density.

4.2.2 Equivalence with Bonemat (T1)

Table 4.6 presents the comparative statistics between Bonemat and SlicerBoneMat for T1.

Table 4.6: Comparative statistics for T1

Statistic	Value
Mean absolute error (MAE)	0.3448 MPa
Root mean square error (RMSE)	0.4212 MPa
Pearson’s correlation coefficient (R)	1.000000
Linear regression slope	1.000000
Linear regression intercept	-0.0043 MPa
Maximum absolute difference	0.9998 MPa

The results demonstrate near-perfect equivalence between the two tools. The Pearson correlation coefficient and linear regression slope are both 1.000000 to six decimal places, and the regression intercept of -0.0043 MPa is negligible relative to the modulus range of the dataset. The MAE of 0.3448 MPa and RMSE of 0.4212 MPa represent

errors of approximately 0.01% of the mean assigned modulus of 3600.9 MPa, which is well within any threshold that would be considered meaningful for finite element analysis of bone. The maximum absolute difference across all elements is 0.9998 MPa, less than 1 MPa across a modulus range of nearly 16000 MPa.

Figure 4.1 presents the per-element scatter plot of SlicerBoneMat against Bonemat moduli for T1. The data points lie along the identity line with no visible deviation across the full modulus range, and the regression line is indistinguishable from the identity line at this scale.

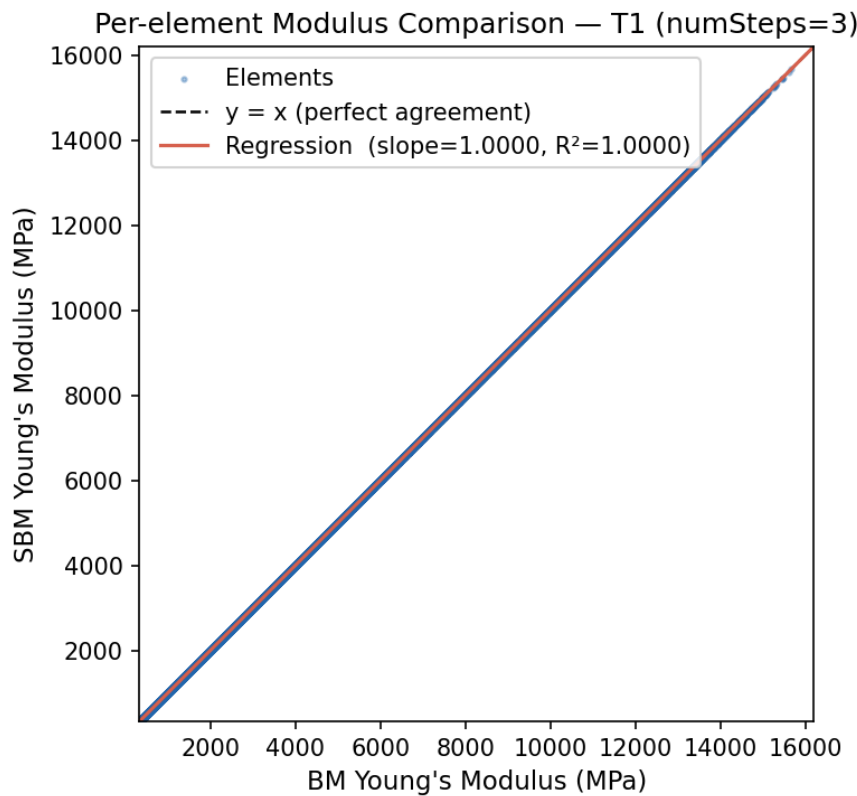


Figure 4.1: Scatterplot of T1

Figure 4.2 presents the residuals plot of SlicerBoneMat minus Bonemat moduli against Bonemat values for T1. The residuals are uniformly distributed around zero across the entire modulus range with no discernible trend, confirming that the small disagreements between the two tools are not concentrated at any particular modulus value (neither at low trabecular nor high cortical values). This is the expected pattern of

a well-behaved numerical implementation, and is strong evidence that SlicerBoneMat reproduces Bonemat’s material assignment without systematic bias.

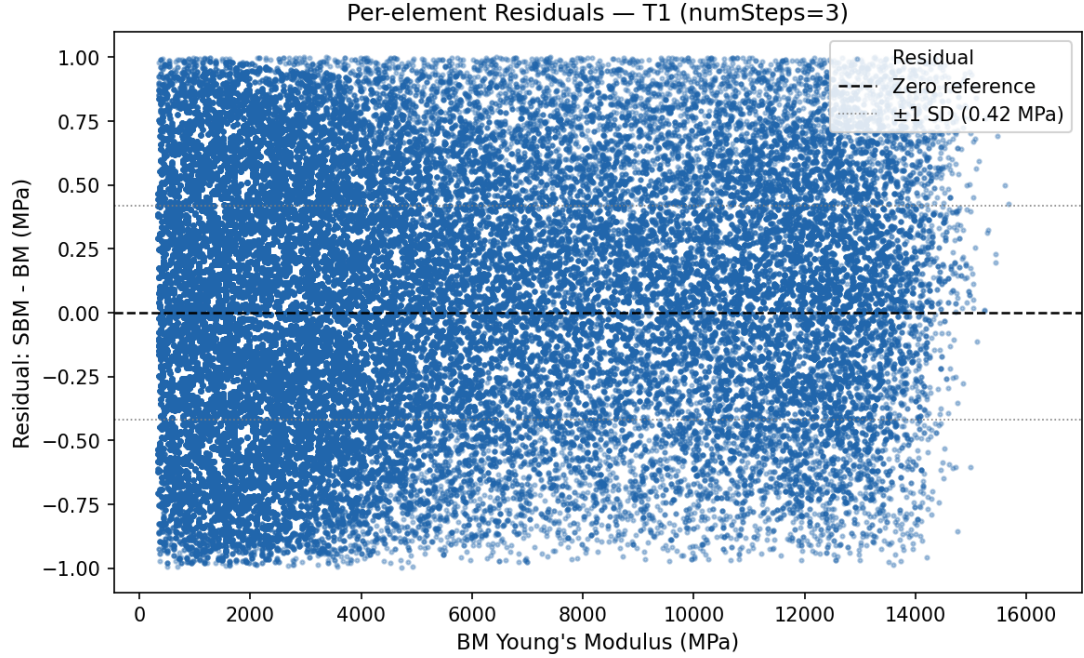


Figure 4.2: Residual scatterplot of T1

4.2.3 Integration Step Sensitivity (T1 vs T2)

Table 4.7 presents the comparative statistics between SlicerBoneMat T1 (*numSteps* = 3) and SlicerBoneMat T2 (*numSteps* = 4).

Table 4.7: Comparative statistics for T1 against T2

Statistic	Value
Mean absolute error (MAE)	39.6578 MPa
Root mean square error (RMSE)	64.1742 MPa
Pearson’s correlation coefficient (R)	0.999855
Linear regression slope	0.991417
Linear regression intercept	19.1659 MPa
Maximum absolute difference	863.0 MPa

Increasing the integration step count from 3 to 4 produces considerably more variation in the assigned moduli than the tool-to-tool difference observed in T1. The MAE

between T1 and T2 is 39.6578 MPa and the RMSE is 64.1742 MPa, approximately two orders of magnitude larger than the equivalent metrics against BoneMat. The maximum absolute difference reaches 863.0 MPa, and the regression slope of 0.991417 departs slightly but measurably from unity. The correlation coefficient remains very high at 0.999855, indicating that the two runs agree strongly on the relative ordering and spatial distribution of material properties, but that the absolute modulus values assigned to individual elements differ non-trivially.

This finding has an important implication: the choice of *numSteps* has a larger effect on the absolute values of the assigned moduli than the choice of tool. A researcher switching from Bonemat to SlicerBoneMat at the same *numSteps* setting can expect near-identical results, but a researcher changing *numSteps* within SlicerBoneMat should be aware that this parameter meaningfully influences the output. This underscores the importance of reporting *numSteps* alongside any results produced by SlicerBoneMat, and of maintaining a consistent *numSteps* value across studies that wish to make direct comparisons.

Figure 4.3 presents a scatter plot of SlicerBoneMat T2 moduli against SlicerBoneMat T1 moduli, with the $y = x$ identity line as the reference. The plot shows close agreement at both ends of the modulus range (the low-modulus trabecular elements and the sparse population of very high-modulus cortical elements both lie near the identity line) but fans out noticeably in the middle to upper-middle of the distribution. This suggests that the integration step change has the greatest influence on elements in the transitional zone between trabecular and cortical bone, where HU values are moderate but gradients within individual elements are likely steepest, making the numerical integration most sensitive to sampling density. The fanning is non-trivial and consistent with the RMSE and maximum absolute difference reported in the above table.

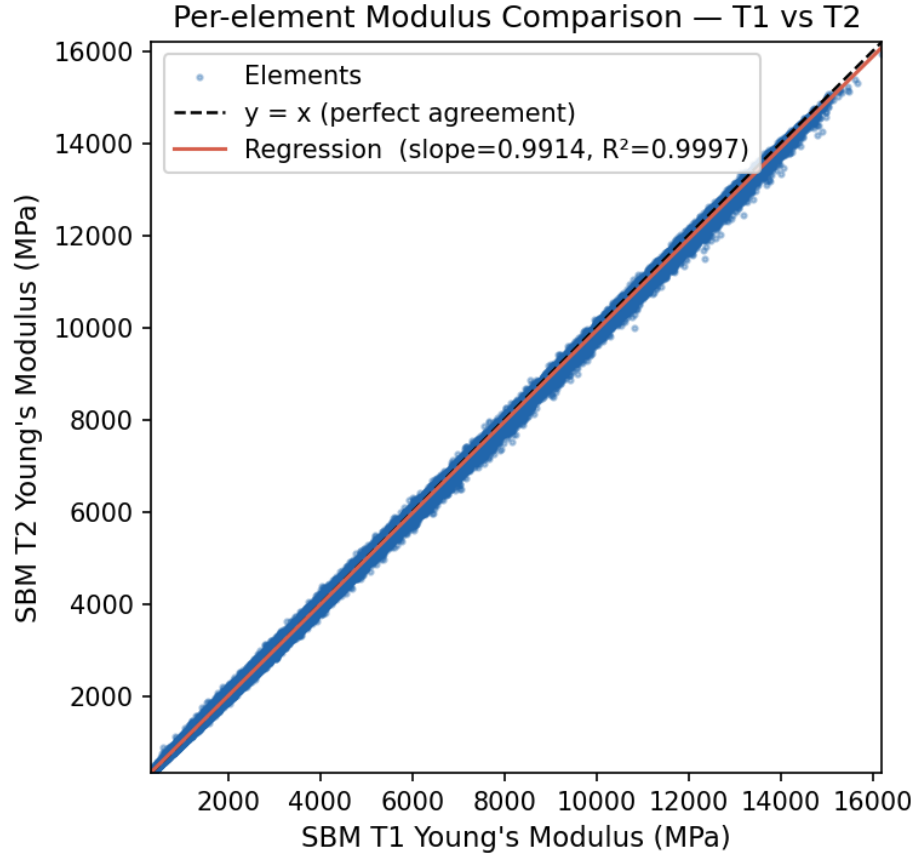


Figure 4.3: Scatterplot of T1 against T2

Figure 4.4 presents the residuals of SlicerBoneMat T2 minus SlicerBoneMat T1, plotted against T1 modulus values. Unlike the T1 residuals against Bonemat, which showed no systematic trend, the T2 minus T1 residuals exhibit a clear and structured pattern. At low modulus values, T2 tends to overestimate relative to T1, producing positive residuals. As modulus increases into the middle and upper-middle of the distribution (the transitional zone identified in the scatter plot) the residuals tend to cross zero and T2 begins to underestimate relative to T1, with the magnitude of the negative residuals growing before slightly contracting again at the highest modulus values where the element population is sparse.

This sign reversal suggests that increasing the integration step count does not uniformly scale the assigned moduli up or down, but rather shifts them in a modulus-dependent

direction, likely reflecting how the denser quadrature grid samples the CT scalar field differently across regions of varying HU gradient character. This structured behaviour reinforces that *numSteps* is a parameter with non-trivial influence on the material field, and that its effect cannot be characterised simply as a constant offset or scaling factor.

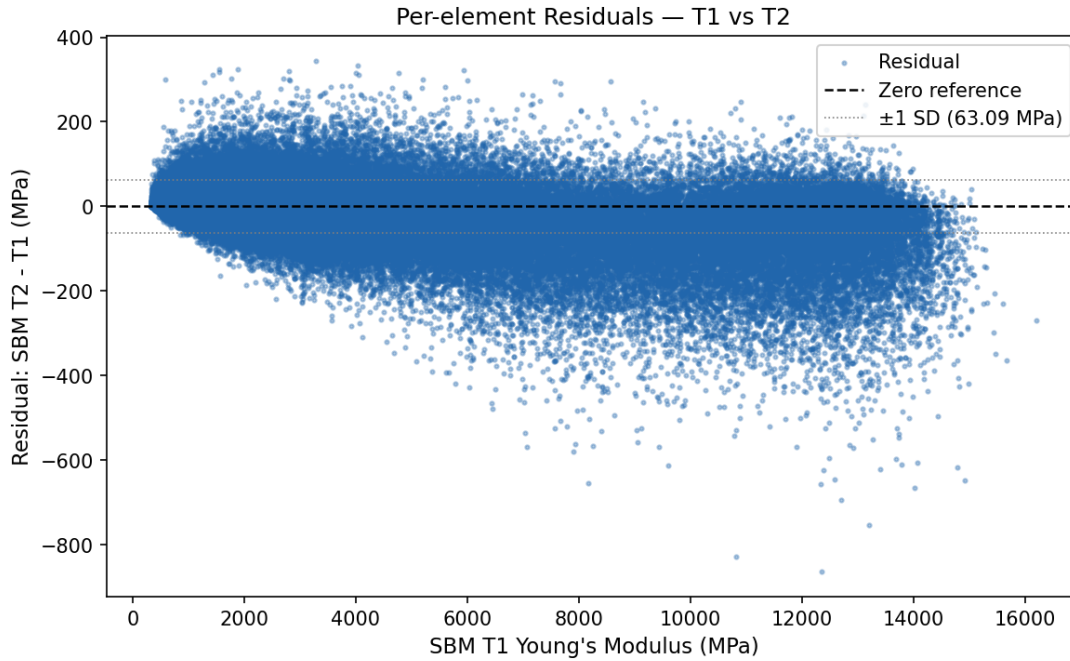


Figure 4.4: Residual scatterplot of T1 against T2

Figure 4.5 presents the three-way histogram overlay of Bonemat, SlicerBoneMat T1, and SlicerBoneMat T2 modulus distributions. The Bonemat and SlicerBoneMat T1 distributions are visually indistinguishable at this scale, appearing as a single overlapping band. The SlicerBoneMat T2 distribution follows the same shape but shows a slight increase in lower-modulus values and a decrease in higher-modulus elements. Despite the difference being potentially difficult to observe, it is consistent with the values observed in the descriptive statistics.

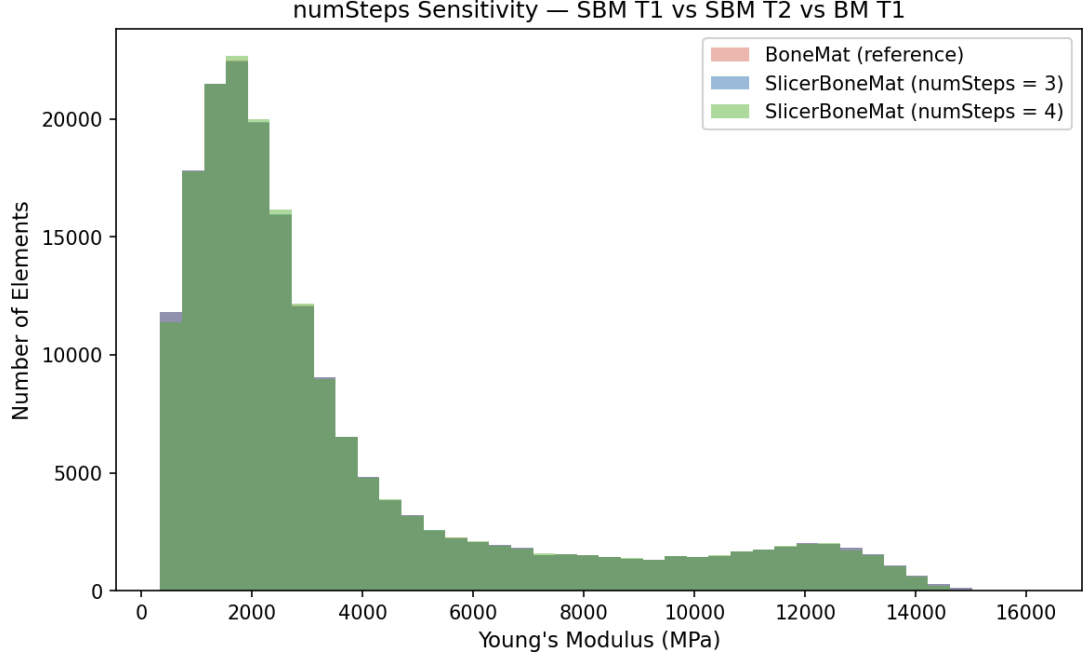


Figure 4.5: Overlapping histogram of T1 and T2

4.3 Material Assignment Accuracy – Full Femur (T3)

Test case T3 applies the equivalence assessment to a full femur specimen at *numSteps* = 3, using CT data sourced from the HFValid dataset with phantom-calibrated conversion coefficients. This test serves two purposes: first, to verify that the equivalence demonstrated on the proximal femur in Section 4.2 generalises to a geometrically larger and more complex specimen; and second, to assess whether the use of a different CT dataset and calibration source introduces any observable effect on the agreement between tools. As noted in Section 4.1.2, the two specimens originate from different CT scans with different calibration coefficients, meaning any differences in equivalence between T1 and T3 may reflect both geometric and dataset-related factors.

4.3.1 Descriptive Statistics

Table 4.8 presents the descriptive statistics of the assigned Young's modulus distributions for Bonemat and SlicerBoneMat on the full femur specimen.

Table 4.8: Descriptive statistics for T3

Statistic (MPa)	Bonemat	SlicerBoneMat
Mean	4518.1	4520.0
Min	334.3	334.0
Max	19752.6	19753.0
Range	19418.4	19419.0
Median	1688.6	1690.0

The two distributions are in close agreement across all reported statistics. The mean moduli differ by only 1.9 MPa, and the minimum values are within 0.3 MPa of each other. The maximum values are nearly identical at 19752.6 MPa and 19753.0 MPa respectively, and the medians differ by just 1.4 MPa. The full femur produces a notably higher mean and maximum modulus than the proximal femur (4518 MPa versus 3601 MPa and approximately 19753 MPa versus 16202 MPa respectively), which is consistent with the inclusion of the full cortical shaft in this specimen, a region of particularly dense bone that is absent from the proximal-only sample.

4.3.2 Equivalence with Bonemat

Table 4.9 presents the comparative statistics between Bonemat and SlicerBoneMat for T3.

Table 4.9: Comparative statistics for T3

Statistic	Value
Mean absolute error (MAE)	2.0652 MPa
Root mean square error (RMSE)	3.5423 MPa
Pearson’s correlation coefficient (R)	1.000000
Linear regression slope	1.000136
Linear regression intercept	1.2795 MPa
Maximum absolute difference	17.6615 MPa

The results confirm strong equivalence between the two tools on the full femur specimen. The Pearson correlation coefficient is 1.000000 and the regression slope is 1.000136, deviating from unity by less than 0.02%. The MAE of 2.0652 MPa and RMSE of

3.5423 MPa are larger in absolute terms than those observed for the proximal femur in T1 (where MAE was 0.3448 MPa and RMSE was 0.4212 MPa), but remain small relative to the modulus range of the dataset. Expressed as a proportion of the mean assigned modulus of 4518 MPa, the RMSE represents an error of approximately 0.08%, which is well within any practically meaningful threshold for bone FE analysis. The maximum absolute difference of 17.6615 MPa, while larger than the sub-1 MPa value observed in T1, similarly represents less than 0.09% of the dataset range.

The modest increase in absolute error metrics relative to T1 is not unexpected. The full femur specimen is geometrically larger, contains a wider range of bone densities, and was calibrated using a different phantom-derived coefficient set. Any one of these factors could contribute to slightly larger per-element disagreements without indicating a systematic failure of equivalence. The regression intercept of 1.2795 MPa, while small, is larger than the -0.0043 MPa observed in T1 and suggests a very slight consistent offset between the two tools on this dataset, though at this magnitude it is unlikely to have any material consequence for FE analysis.

Figure 4.6 presents the per-element scatter plot of SlicerBoneMat against Bonemat moduli for T3. Points lie closely along the $y = x$ identity line across the full modulus range, and the regression line is visually indistinguishable from the identity line, consistent with the near-unity slope and intercept values reported above. The agreement is strong, though a careful inspection of the scatter reveals marginally more spread than was visible in the T1 plot, particularly at higher modulus values, consistent with the higher absolute error metrics.

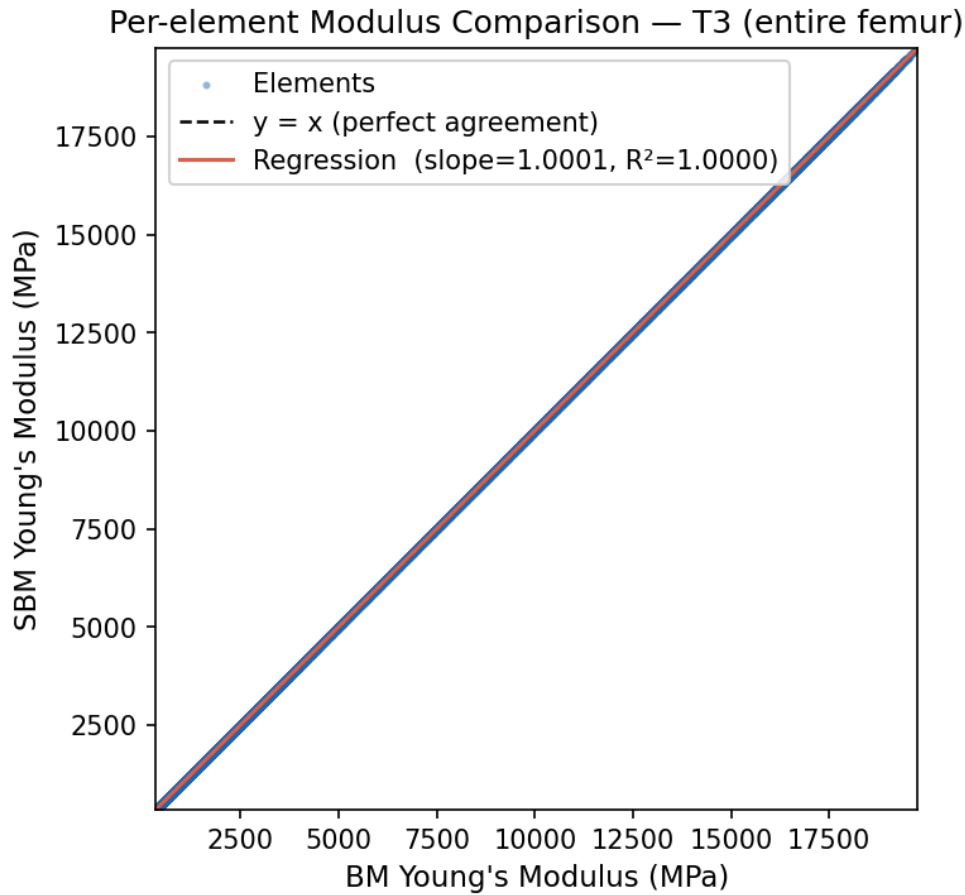


Figure 4.6: Scatterplot of T3

Figure 4.7 presents the residuals plot for T3. The distribution of residuals differs qualitatively from the uniform scatter observed in T1. While the majority of elements fall within a ± 1 MPa band around zero, indicating close agreement for most of the mesh, a notable secondary population of points shows SlicerBoneMat overestimating Bone-mat by a consistently positive margin. This overestimation is present from the lower end of the modulus range at approximately 7 MPa above zero and grows gradually with increasing modulus, reaching the maximum absolute difference of approximately 17 MPa at the upper end of the distribution. This structured overestimation pattern, absent from the T1 residuals, suggests a systematic but small positive bias in Slicer-BoneMat’s assignment relative to Bonemat on this specimen. Whether this reflects a dataset-specific effect (such as the phantom calibration coefficients affecting the mate-

rial mapping pipeline) or a geometric effect of the larger mesh is not determinable from this test alone, and is discussed further in Section 5.

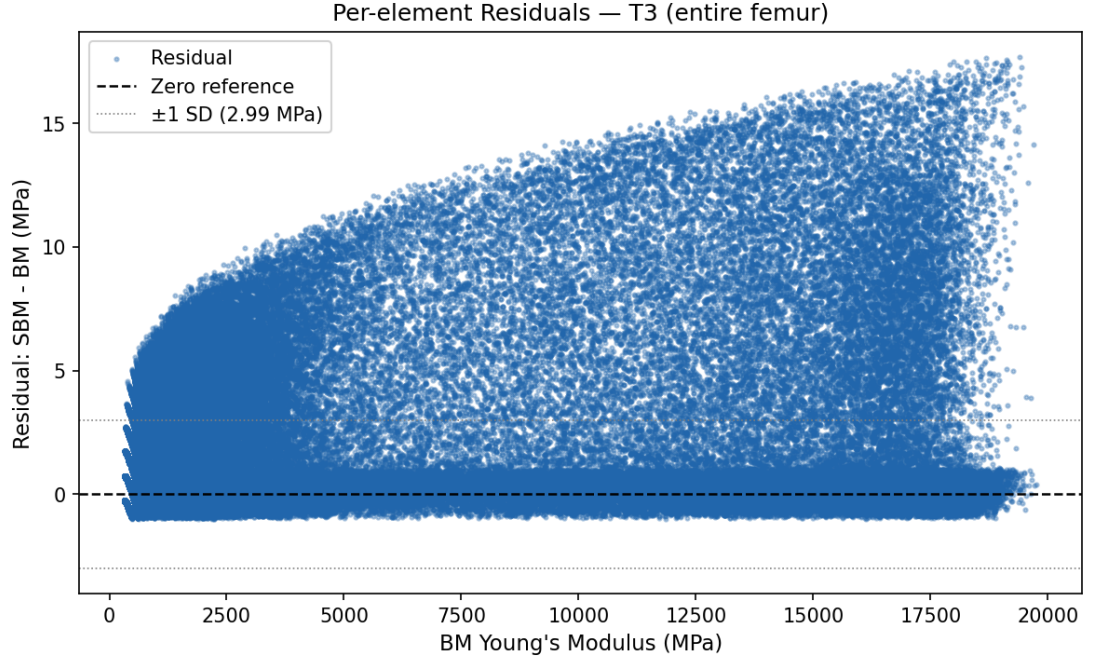
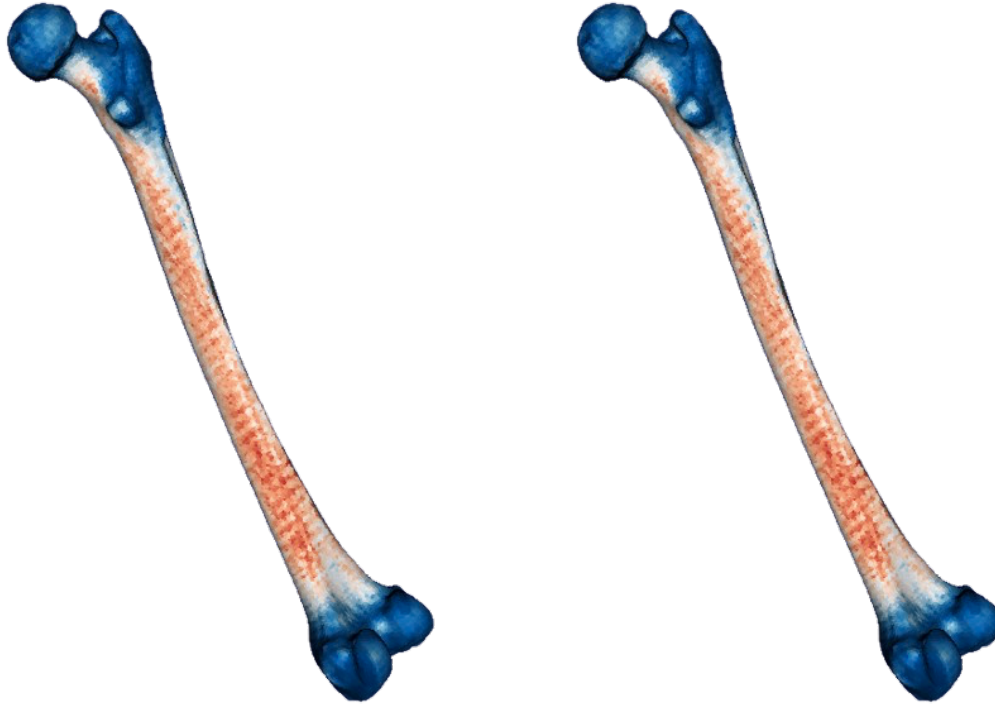


Figure 4.7: Residual scatterplot of T3

Figure 4.8 presents screenshots of the full femur mesh rendered in SlicerBoneMat’s 3D viewport, coloured by assigned Young’s modulus. The characteristic spatial distribution of bone material properties is clearly visible: the cortical shell of the femoral shaft appears as a red high-modulus band, while both the femoral head and the distal condyles present as predominantly blue low-modulus regions consistent with their largely trabecular composition, with the transition between cortical and trabecular zones visible as a gradient along the shaft. This spatial distribution is qualitatively consistent with the expected material heterogeneity of a human femur and provides visual confirmation that the mapping has produced a physically plausible result across the full geometry.



(a) Bonemat full femur

(b) SlicerBoneMat full femur

Figure 4.8: Qualitative comparison of material assignment on full femur

4.4 Binning Accuracy (T4)

Test case T4 evaluates the accuracy of SlicerBoneMat’s modulus binning implementation against Bonemat, using the proximal femur specimen with a gap value of 50 MPa. As described in Section 3.5, binning groups of nearby modulus values into shared bins reduces the number of distinct material definitions in the output file. This test assesses whether SlicerBoneMat’s modified binning implementation produces bin structures equivalent to Bonemat’s, and whether the assignment of elements to those bins is consistent between the two tools.

Both SlicerBoneMat and Bonemat were configured to use the maximum grouping density, whereby each bin is assigned the highest modulus value within its group rather than the mean. This choice is consistent across both tools and represents the more

conservative option from a structural standpoint, avoiding any underestimation of local stiffness that would result from mean-based grouping.

4.4.1 Binning Statistics

Table 4.10 presents the binning statistics for T4.

Table 4.10: Binning statistics

Metric	Value
Unique modulus values (SBM)	300
Unique modulus values (BM)	300
Mean intra-bin spread (SBM)	49.0300 MPa
Mean intra-bin spread (BM)	48.9498 MPa
MAE on binned values	0.0000 MPa
RMSE on binned values	0.0000 MPa
Bin match rate (tolerance $< 10^{-6}$)	98.48%
Bin match rate (tolerance $< 2 \times 10^{-6}$)	100.00%

Both tools produced exactly 300 unique modulus values from the proximal femur mesh, indicating that the binning algorithm partitioned the modulus range into the same number of groups. The mean intra-bin spread (the average range of raw modulus values collapsed into a single bin) is 49.03 MPa for SlicerBoneMat and 48.95 MPa for Bonemat, a difference of less than 0.1 MPa. This close agreement in spread confirms that the two tools are grouping elements at effectively the same resolution, consistent with the shared gap value of 50 MPa.

The MAE and RMSE on the binned values both register as 0.0000 MPa to four decimal places, indicating that the per-element differences between the two tools' bin assignments are below the threshold of reportable precision at this rounding level. The tolerance-based match rate analysis confirms this: 98.48% of elements have bin assignments within 1×10^{-6} MPa of each other, rising to 100% within 2×10^{-6} MPa. The 1.52% of elements falling outside the tighter threshold correspond to elements whose raw modulus values sit close to a bin boundary, where sub-micron floating point differences in the two implementations cause them to round to fractionally different values

before converging within the wider tolerance. This is a well-understood numerical artefact rather than an algorithmic disagreement, and at a magnitude of less than 2×10^{-6} MPa it has no practical consequence for finite element analysis.

4.4.2 Bin Assignment Residuals

Figure 4.9 presents the per-element residuals of SlicerBoneMat minus Bonemat binned moduli, plotted against Bonemat values, with the y-axis fixed at ± 3 MPa to contextualise the residuals against the broader modulus range.

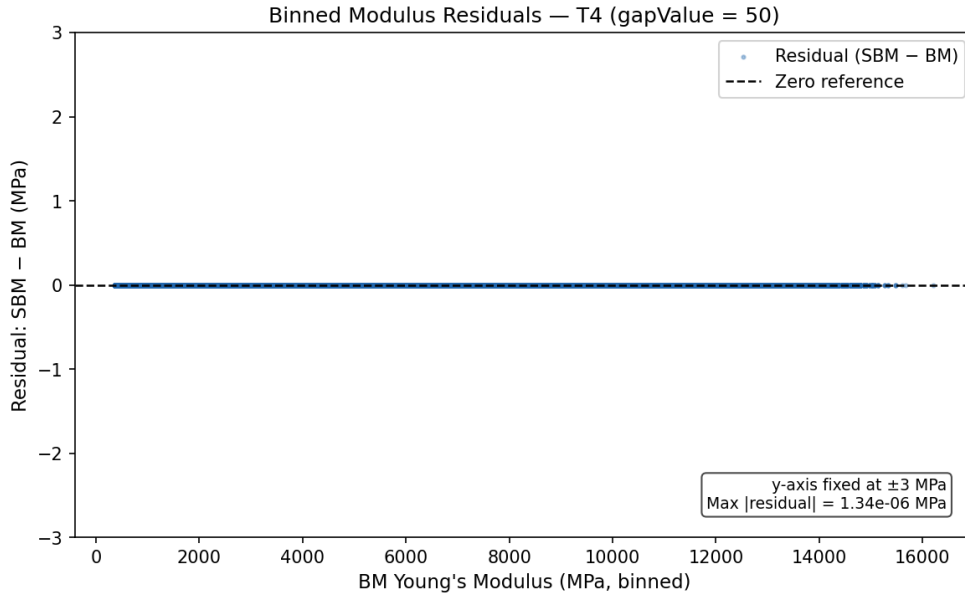


Figure 4.9: Residual scatterplot of T4

All 300 data points lie within 2×10^{-6} MPa of zero, appearing as a flat line along the zero reference at this scale. A marginal upward trend in residual magnitude with increasing modulus value is detectable in the underlying data, consistent with the expected behaviour of floating point arithmetic where relative precision degrades slightly with larger numbers. This trend is not visible at the scale of the figure above, where all points appear as a flat line along the zero reference, reflecting the negligible magnitude of the disagreement, particularly in contrast to the structured residual patterns observed in Sections 4.2 and 4.3.

4.4.3 Effect of Binning on the Modulus Distribution

Figure 4.10 presents a side-by-side comparison of the SlicerBoneMat modulus distributions before binning (T1, *gapValue* = 1) and after binning (T4, *gapValue* = 50), using 80 histogram columns in each panel.

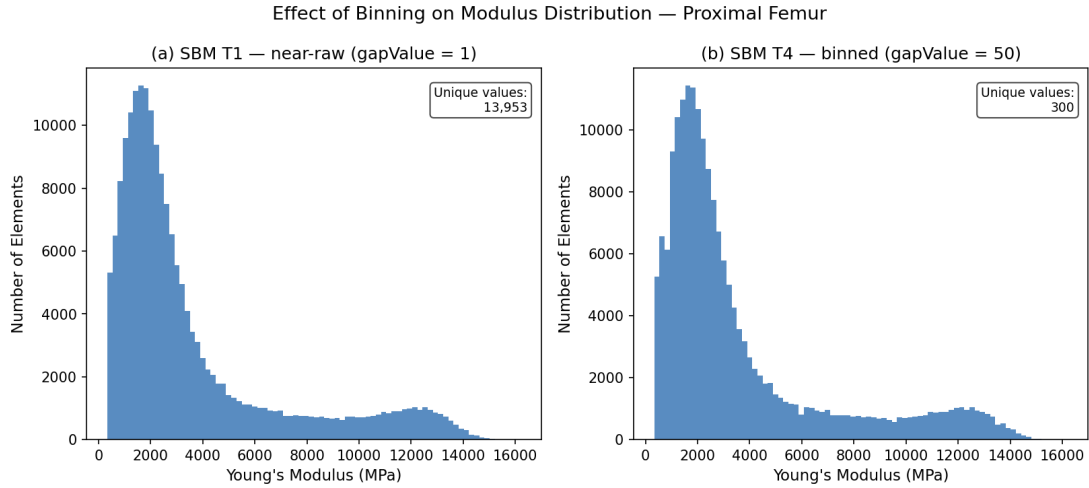


Figure 4.10: Comparison of modulus distribution for T1 against T4

The overall shape of the distribution is preserved after binning, as the characteristic right-skewed profile with a high concentration of low-modulus trabecular elements and a long upper tail of cortical bone values is clearly visible in both panels. The primary effect of binning is a consolidation of the continuous modulus field into more discrete spikes, where elements that previously held closely-spaced but distinct values are collapsed into shared bin values, producing localised concentrations in the binned distribution.

It should be noted that the appearance of the binned panel is moderately sensitive to the number of histogram columns used for display: at different column counts the spike structure changes in its visual prominence, as individual bins either align with or straddle column boundaries. This is a visualisation artefact of representing a discretised distribution as a histogram and does not reflect any underlying change in the data. The 80-column presentation shown here was selected as the count at which the binned and raw distributions are most visually comparable.

4.5 Computational Performance (T5)

Two computational performance tests are conducted to characterise how SlicerBoneMat’s runtime scales with the two parameters most likely to influence it: mesh size and integration step count. Both tests were run on the proximal femur CT volume with identical density-to-modulus parameters, a gap value of 1, and a minimum modulus of 10 MPa, isolating the variable of interest in each case. Bonemat’s runtime is not benchmarked as a direct comparison given the emulated Windows environment described in Section 4.1.4; informal observation during testing confirmed that Bonemat completed all test cases in under five seconds, and runtime comparison between the two tools is therefore discussed qualitatively in Section 5.

4.5.1 Computation Time vs Mesh Size (T5a)

To generate meshes of varying size from the same proximal femur geometry, the feature sampling parameter of the Cleaver mesher (available within the SlicerSegmentMesher extension) was progressively decreased, producing finer discretisations of the same surface. Decreasing the feature sampling parameter increases the density of the tetrahedral mesh, yielding a larger number of smaller elements. All meshes were generated from the same CT segmentation, ensuring that geometry and CT volume were held constant across the test and that only element count varied. The number of integration steps was held constant at $numSteps = 3$ for all runs.

Figure 4.11 presents the computation time as a function of element count, with a linear line of best fit overlaid. The relationship between element count and computation time is strongly linear, with the fit equation and R^2 value annotated on the figure. This is the expected theoretical behaviour: each element is processed independently by the mapping engine with no inter-element communication or data dependency, meaning the total runtime scales directly with the number of elements. The confirmed linearity has the practical implication for users that given the slope of the fitted line, the expected runtime for a mesh of any size can be estimated directly by linear extrapolation from

the reported equation, without needing to run a calibration test on their own hardware. In particular, the slope of 0.000181 implies that every increment of 10000 in the number of mesh elements increases the computation time by approximately 1.81 seconds.

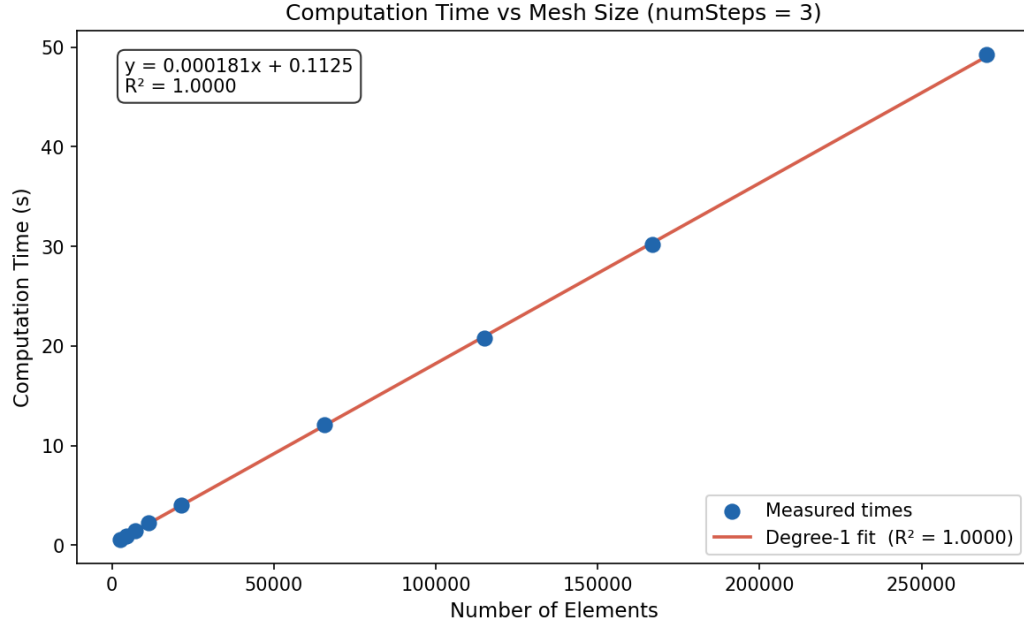


Figure 4.11: Scatterplot of T5a

4.5.2 Computation Time vs Integration Steps (T5b)

The integration step sensitivity test was conducted on a fixed mesh of 115,116 elements, with *numSteps* varied across a range of values while all other parameters were held constant. Using a fixed mesh ensures that any change in computation time is attributable solely to the change in quadrature density rather than mesh size.

Figure 4.12 presents the computation time as a function of *numSteps*, with a cubic polynomial line of best fit overlaid. The relationship between *numSteps* and computation time follows a cubic trend, consistent with the theoretical complexity of the uniform quadrature scheme. As described in Section 3.4.4, the number of integration points within each tetrahedral element scales as $O(n^3)$ with *numSteps*, since the quadrature grid is sampled in three nested loops across the element's parametric space. The cubic fit to the measured data confirms that this theoretical complexity is reflected in the

observed runtime, validating that the implementation scales as expected and that no unexpected overhead is introduced at higher integration densities. The practical consequence is that increasing *numSteps* carries a notable computational cost (doubling *numSteps* increases runtime by approximately eightfold), and users should therefore select the lowest *numSteps* value consistent with acceptable accuracy, informed by the sensitivity analysis presented in Section 4.2.3.

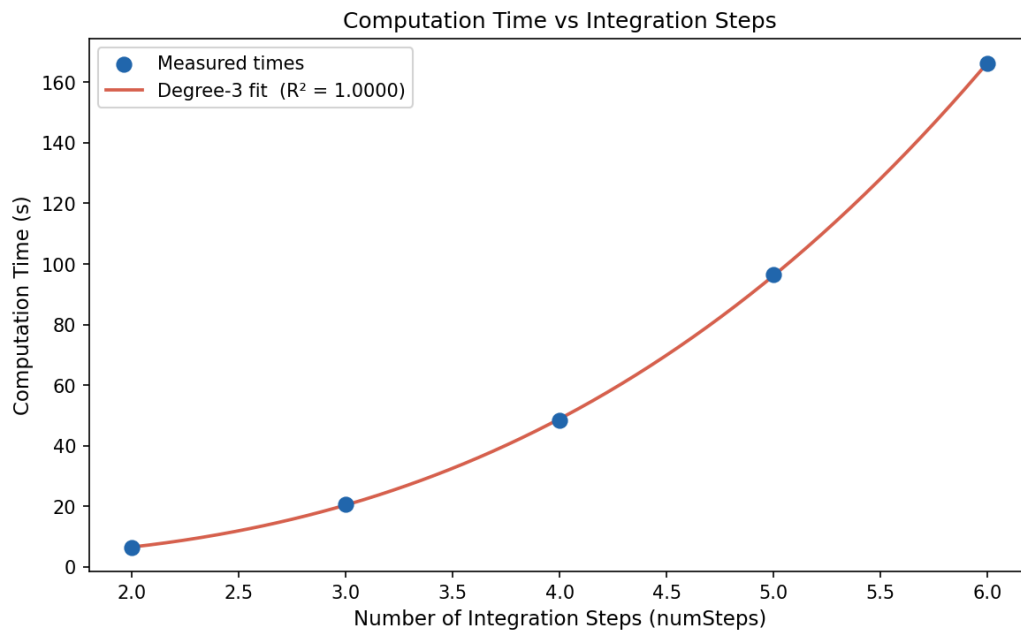


Figure 4.12: Scatterplot of T5b

4.6 Mesh Statistics Accuracy (T6)

This test case evaluates whether the mesh quality statistics computed by SlicerBoneMat agree with those reported by FEBio Studio for the same proximal femur mesh. As described in Section 3.2.6, SlicerBoneMat computes four per-element statistics: element volume, tetrahedral quality (radius-edge ratio), minimum edge length, and maximum edge length. FEBio Studio reports the same four metrics, making a direct comparison possible. This test validates that SlicerBoneMat provides mesh diagnostics that are consistent with the industry-standard biomechanics FE environment, giving users confidence that the quality information presented prior to mapping is reliable.

4.6.1 Summary Statistics

Table 4.11 presents the minimum, maximum, and mean values for each of the four statistics, reported by both tools to four decimal places, which is the maximum precision available from FEBio Studio’s output.

Table 4.11: Mesh statistics for SBM vs FEBio Studio

Statistic	Metric	SBM	FEBio Studio	Agreement
Element volume	Min	0.0050	0.0050	✓ ^a
	Max	13.0032	13.0032	✓ ^a
	Mean	2.9606	2.9606	✓ ^a
Tet quality	Min	0.6217	0.6217	✓ ^a
	Max	4.0228	4.0228	✓ ^a
	Mean	1.2872	1.2872	✓ ^a
Minimum element edge length	Min	0.3145	0.3145	✓ ^a
	Max	4.3945	4.3945	✓ ^a
	Mean	2.0617	2.0617	✓ ^a
Maximum element edge length	Min	0.8488	0.8488	✓ ^a
	Max	6.4062	6.4062	✓ ^a
	Mean	4.2480	4.2480	✓ ^a

^a Exact agreement at 4 decimal places. Any underlying difference is smaller than 5×10^{-5} in the relevant units.

FEBio Studio reports mesh statistics to 4 decimal places, which defines the resolution at which any disagreement between the two tools could be detected. Since all reported values agree exactly at that resolution, any underlying difference between the two implementations must be smaller than 0.00005 in the relevant units (the maximum undetectable discrepancy given 4 d.p. rounding).

Whether that represents a true implementation difference or genuine equivalence cannot be determined from this comparison alone, but the result is that the two tools are indistinguishable at the precision FEBio Studio makes available; every minimum, maximum, and mean statistic matches at the resolution available for comparison. The minimum tetrahedral quality value of 0.6217 is consistent with the theoretical minimum of approximately 0.612 for an equilateral tetrahedron, as mentioned in Section 3.2.6, confirming that the best-quality elements in the mesh approach the ideal shape.

The maximum quality value of 4.0228 indicates the presence of some degenerate elements, though a mean of 1.2872 suggests the overall mesh quality is reasonable for biomechanical FE analysis.

4.6.2 Distribution Comparison

Figure 4.13 presents the distribution of each statistic from both tools as a 4×2 grid of histograms, with SlicerBoneMat in the left column and FEBio Studio in the right column for each metric.

The distributions are visually indistinguishable across all four metrics. The element volume distribution is heavily right-skewed, with the majority of elements concentrated at small volumes and a long tail of larger elements. The radius-edge ratio distribution is likewise right-skewed, concentrated near the theoretical minimum with a tail extending toward higher values, reflecting a mesh dominated by well-shaped elements with a small population of more degenerate ones. The maximum edge length distribution shows an approximately unimodal profile centred near its mean value, while the minimum edge length distribution is notably bimodal; the origin of this bimodality is not immediately apparent from the mesh statistics alone and warrants further inspection if mesh refinement is considered.

In each case the FEBio Studio histogram is a close visual match to the SlicerBoneMat equivalent, with no systematic differences in shape, peak location, or tail behaviour detectable at this resolution. Combined with the exact summary statistic agreement in the above table, this confirms that SlicerBoneMat’s mesh statistics panel is a fully reliable, albeit minimal, substitute for FEBio Studio’s equivalent functionality for the purposes of pre-mapping mesh quality assessment.

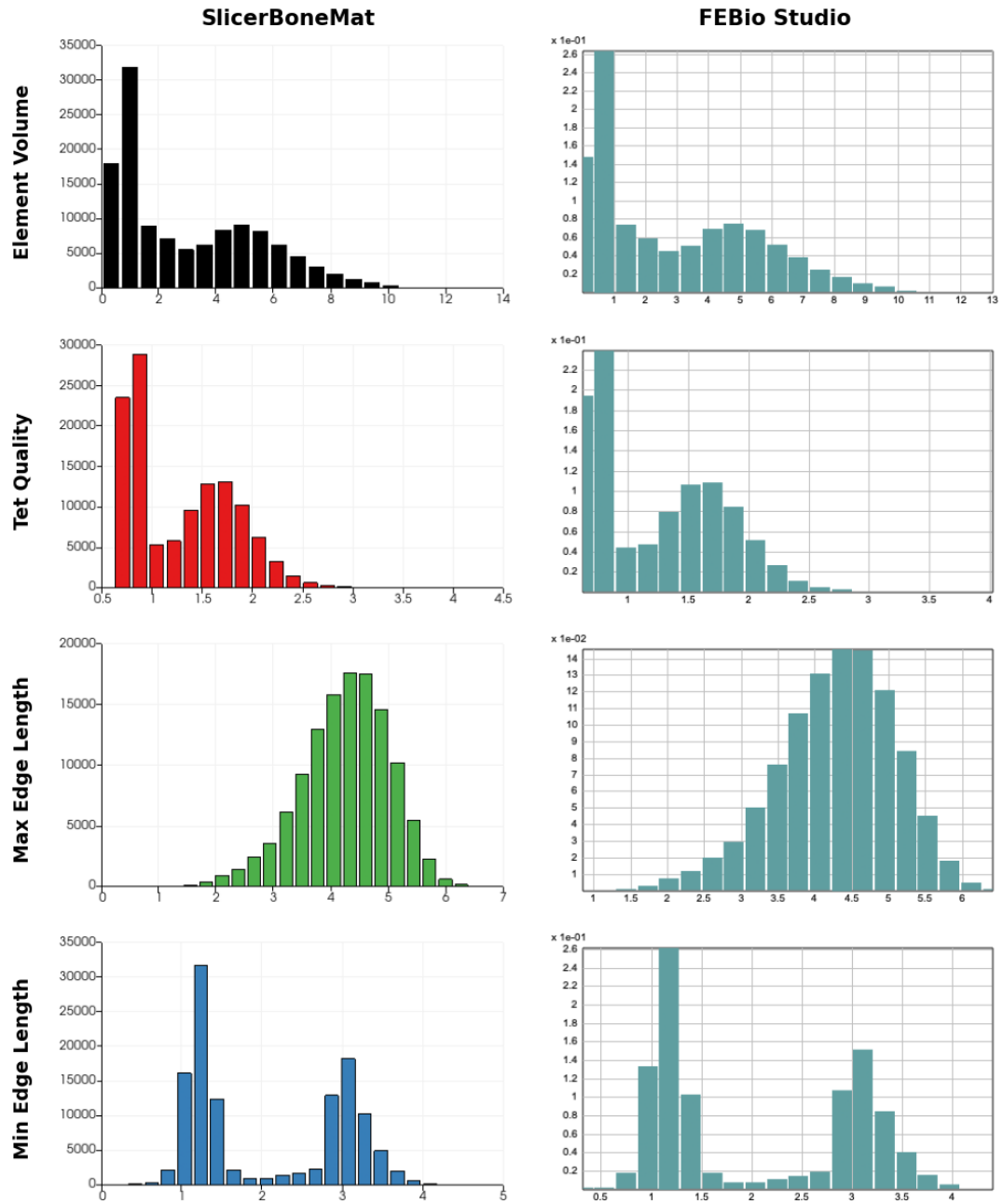


Figure 4.13: Mesh statistic distribution grid for SBM against FEBio Studio

4.7 Summary of Validation Results

Table 4.12 summarises the headline metrics from each test case conducted in this validation study.

Table 4.12: Mesh statistics for SBM vs FEBio Studio

Test	Description	Key Metric	Value
T1	Proximal femur equivalence	RMSE vs Bonemat	0.4212 MPa
T2	Integration step sensitivity	RMSE vs SBM T1	64.1742 MPa
T3	Full femur equivalence	RMSE vs Bonemat	3.5423 MPa
T4	Binning accuracy (<i>gapValue</i> = 50)	Match rate ^a	100%
T5a	Computation time vs mesh size	Linear fit R^2	$R^2 = 1.0000$
T5b	Computation time vs <i>numSteps</i>	Scaling behaviour	Cubic, $O(n^3)$
T6	Mesh statistics vs FEBio Studio	Agreement at 4 d.p.	Exact

^a At 2×10^{-6} tolerance

Across all test cases, SlicerBoneMat produces results consistent with its design objectives. The equivalence tests demonstrate that the extension replicates Bonemat’s material assignment to a high degree of accuracy on both a proximal and full femur specimen, with RMSE values representing approximately 0.01% and 0.08% of their respective datasets’ average modulus. The binning implementation is functionally identical to Bonemat’s at any practically meaningful tolerance, and mesh quality statistics are indistinguishable from FEBio Studio’s output at the available comparison precision. Computational performance scales as theoretically expected with both mesh size and integration density. A significant finding across the study is that the choice of *numSteps* has a substantially larger influence on absolute modulus values than the choice of tool, with the within-SlicerBoneMat RMSE between *numSteps* = 3 and *numSteps* = 4 being approximately 152 times larger than the tool-to-tool RMSE at *numSteps* = 3. This underscores that consistent reporting of integration parameters is essential for reproducibility across studies using either tool.

Chapter 5

Discussion

5.1 Equivalence with Bonemat

The primary objective of this project was to implement a material mapping extension for 3D Slicer that reproduces the output of Bonemat with sufficient fidelity to serve as a practical replacement in bone FE modelling workflows. The equivalence results from T1 and T3 support this conclusion. All equivalence tests were conducted using the V3 integration algorithm, consistent with Bonemat’s default setting. This choice is motivated by Taddei et al. [TSH⁺07], who demonstrated that V3 (which integrates the Young’s modulus field directly rather than the density field) produces lower average and peak strain prediction errors than V2, and using the same algorithm version as Bonemat’s default is a prerequisite for a meaningful equivalence comparison.

For the proximal femur at *numSteps* = 3, SlicerBoneMat produced per-element Young’s modulus values with an RMSE of 0.4212 MPa against Bonemat, which represents less than 0.01% of the mean assigned modulus of 3600.9 MPa. The Pearson correlation coefficient and regression slope were both 1.000000 to six decimal places, and the residuals showed no systematic trend across the full modulus range. For the full femur at *numSteps* = 3, the RMSE was 3.5423 MPa, less than 0.08% of the mean assigned modulus of 4518.1 MPa, with an equally high correlation coefficient. By practical validation

thresholds, these results constitute equivalence. Studies validating in-house material mapping implementations against Bonemat have generally reported agreement criteria in terms of correlation coefficients and regression slopes rather than absolute RMSE thresholds, consistently accepting $R^2 > 0.99$ and slopes within 2% of unity as evidence of equivalence [PG16]. SlicerBoneMat meets these criteria on both specimens.

The modest increase in absolute error metrics between T1 and T3 requires some interpretation. Three factors differentiate the full femur test from the proximal femur test: a larger and geometrically more complex mesh, a different CT dataset sourced from a different patient, and a different calibration coefficient set derived from phantom measurements rather than internal tissue reference points. Any of these could contribute to the slightly larger per-element disagreements observed. The structured overestimation pattern visible in the T3 residuals, where SlicerBoneMat consistently assigns moduli approximately 7 to 17 MPa above Bonemat with increasing modulus value, may point to a calibration-related effect. The HFValid phantom calibration coefficients used for the full femur define a specific linear HU-to-density relationship that differs from the internal tissue calibration used for the proximal femur. Small differences in how the two tools handle the floating point arithmetic of this calibration chain (particularly across a wider modulus range in a larger mesh) could accumulate into a consistent positive offset of the magnitude observed. Disentangling these contributing factors would require running both specimens through matched calibration protocols, which was not possible given the dataset constraints described in Section 4.1.2. This remains an open question for future validation work.

Even so, the overestimation is small in absolute terms and does not change the overall equivalence finding. A difference of 17 MPa at the upper end of a modulus range spanning nearly 20,000 MPa represents a relative error of less than 0.09%, which is well within the uncertainty introduced by the choice of density-elasticity relationship itself, a source of inter-study variability that the literature has documented to be orders of magnitude larger [HPS⁺08, MBK03].

It is worth noting that the validation presented here is comparative rather than abso-

lute, validating that SlicerBoneMat replicates Bonemat’s output, not that Bonemat’s output is itself accurate relative to experimental ground truth. However, this distinction does not weaken the value of the equivalence result. Bonemat has been validated against experimental mechanical testing data in the original publications by Taddei et al. [TSH⁺07, THP⁺08], which demonstrated that the V3 integration algorithm produces strain predictions in close agreement with measured values on cadaveric femur specimens. By establishing equivalence with Bonemat, SlicerBoneMat draws on this validation indirectly, and hence researchers using SlicerBoneMat with the same parameters can reasonably expect the same accuracy relative to physical measurements that Bonemat has been shown to achieve.

5.2 The Influence of *numSteps*

The sensitivity analysis showed that increasing *numSteps* from 3 to 4 on the proximal femur produces an RMSE of 64.1742 MPa between the two SlicerBoneMat runs, approximately 152 times larger than the tool-to-tool RMSE of 0.4212 MPa observed in T1. This has a practical implication: a researcher switching from Bonemat to SlicerBoneMat at the same *numSteps* setting can expect closely matched results, but a researcher changing *numSteps* within either tool introduces variation substantially larger than the difference between the two implementations.

To the author’s knowledge, this sensitivity has not been explicitly characterised in the published literature for this class of material mapping software. Bonemat’s documentation recommends a default integration step count but does not quantify the sensitivity of the output to this parameter. The present study provides a quantitative characterisation of this sensitivity for a BoneMat-equivalent implementation, and the finding supports consistent reporting of *numSteps* alongside any published material mapping results, analogous to the established practice of reporting the choice of density-elasticity relationship.

The residuals plot for T2 revealed that the sensitivity to *numSteps* is not uniform across

the modulus range but is concentrated in the middle to upper-middle of the distribution, where HU gradients within individual elements are likely steepest. This is consistent with the expected behaviour of the uniform quadrature scheme: in regions where the CT scalar field varies steeply within an element volume, a denser quadrature grid samples the gradient more accurately, yielding a different integrated modulus than a coarser grid. At very low moduli (trabecular bone with low and relatively uniform HU values) and at the high-modulus extremes (cortical bone with sparse element populations), the sensitivity is lower, either because the HU field is more homogeneous within elements or because the element population is too small to dominate the aggregate statistics.

The computation time results from T5b showed cubic scaling in runtime with *numSteps*, with a cubic fit of $R^2 = 1.0000$. This means that doubling *numSteps* increases computation time by a factor of approximately eight. Given that the T1 results already demonstrate strong equivalence with Bonemat at *numSteps* = 3, and given the cubic cost of increasing this parameter, *numSteps* = 3 is a reasonable default for femoral bone mapping at the mesh resolutions tested here. Users requiring higher integration accuracy for meshes with very fine element sizes relative to CT voxel dimensions may wish to increase this value, but should be aware of both the computational cost and the distributional effect documented in T2.

5.3 Positioning Against Existing Tools

The validation results establish that SlicerBoneMat is more than a direct reimplementa-tion of Bonemat; it also addresses several workflow limitations that matter for modern research workflows.

Compared to Bonemat, SlicerBoneMat addresses several major limitations identified in Section 2.3. It runs natively on Windows, macOS, and Linux, removing the platform restriction that excludes many members of the research community. It is fully open-source, distributed under the GNU General Public Licence (GPL), enabling researchers to inspect, modify, and build upon the implementation with the requirement that any

derivative works are similarly distributed as open-source under the same licence, ensuring the tool and its extensions remain accessible to the research community. It exposes a Python API through *BoneMatLogic*, enabling scripted batch processing and programmatic integration into automated pipelines, which Bonemat’s lack of any API makes impossible. It provides immediate in-application visualisation of the mapped material field, allowing quality assurance without exporting to a separate tool. Its preset system allows named parameter configurations to be saved and reloaded across sessions, replacing Bonemat’s manual parameter file editing with a streamlined interface. It produces output in four formats (VTK, Abaqus, Ansys, and FEBio), with FEBio exports adding support beyond Bonemat’s existing formats, enabling direct export to the leading open-source biomechanics solver without any manual file editing or format conversion. The material mapping accuracy demonstrated in T1 and T3 confirms that these workflow improvements do not appear to reduce computational fidelity.

Compared to MITK-GEM, SlicerBoneMat’s main advantage is its validation evidence. As noted in Section 2.4, MITK-GEM explicitly states that its material mapping has not been validated against in-vitro strength measurements or cohort databases. The validation study presented in Chapter 4 provides exactly this kind of quantitative equivalence evidence for SlicerBoneMat, against a widely accepted reference standard. SlicerBoneMat also benefits from the 3D Slicer platform’s broader adoption, more active development community, and more accessible extension architecture compared to the MITK Workbench environment, lowering the barrier to adoption for researchers who are not already MITK developers.

As a companion to FEBio Studio, SlicerBoneMat addresses a major remaining gap in the free, open-source bone FE modelling pipeline. As established in Section 2.7, FEBio Studio is the preferred open-source FE solver environment in biomechanics research, but does not provide a Bonemat-style CT-derived material mapping pathway. The FEBio exporter implemented in SlicerBoneMat produces *.feb* files that FEBio Studio can open and execute directly, without manual editing or format conversion. The mesh statistics validation in T6 demonstrated that SlicerBoneMat’s mesh quality diagnostics matched FEBio Studio’s equivalent output at the available comparison precision, mean-

ing that a researcher can perform all important pre-processing quality checks within SlicerBoneMat with confidence that the results are consistent with what FEBio Studio would report. Together, 3D Slicer with SlicerBoneMat and FEBio Studio provide a complete, free, open-source, cross-platform pipeline from CT image to biomechanical simulation, a combination that was not previously achievable without manual material mapping effort within FEBio Studio, or a commercial solver licence.

5.4 Limitations

Several limitations of the present work are acknowledged here.

Validation scope. The equivalence validation was conducted on two femur specimens only, both from the same anatomical region and element type. While the proximal and full femur results both demonstrate strong agreement with Bonemat, generalisation to other anatomical sites (e.g. vertebrae, tibiae, radii) requires further testing. The femur was chosen as the validation specimen because it is the most extensively studied bone in the CT-based FE literature and the anatomical site for which Bonemat’s own validation was originally conducted, making it the most meaningful single test case for establishing equivalence with the reference implementation. Validation on the femur therefore provides strong evidence of equivalence at the site where the tool is most likely to be used in practice. Similarly, whilst SlicerBoneMat supports quadratic tetrahedral, wedge, and hexahedral elements, these element types were not included in the validation study, and their mapping accuracy against Bonemat has not yet been tested.

Calibration inconsistency between specimens. The two test specimens used different calibration approaches: the proximal femur used an internal tissue-based calibration derived from maximum and minimum HU reference points, while the full femur used phantom-derived coefficients from the HFValid dataset. This inconsistency limits the comparability of the two equivalence results and means the proximal femur test cannot be considered a fully controlled phantom-calibration validation. Future validation should use matched CT and phantom data for all specimens.

CT axis-alignment requirement. SlicerBoneMat requires the CT volume to be axis-aligned in RAS space and raises an error for obliquely oriented volumes, directing the user to resample using Slicer’s existing tools. Whilst this is a reasonable design decision that avoids introducing additional interpolation errors, it represents an additional workflow step for datasets acquired with non-standard orientations.

Single density-elasticity relationship. All validation tests used a single published density-elasticity relationship from Morgan et al. [MBK03], specifically the femoral neck formulation. The accuracy of the material assignment relative to true mechanical properties, as opposed to relative to Bonemat’s output, depends on the validity of this relationship for the specimens tested, a question that is outside the scope of this comparative validation study.

5.5 Future Work

Several directions for future development arise from the limitations identified above. Validation on additional anatomical sites and element types would test the generalisability of the equivalence results and confirm that the extended element type support functions correctly in practice. Cortical-specific mapping, as implemented in Bonemat V4, would align SlicerBoneMat more closely with Bonemat’s current competencies and is a natural next implementation step. Support for quantitative CT calibration protocols, where calibrated bone-mineral-density values are embedded in the DICOM metadata rather than derived from a separate phantom scan, would reduce the current dependency on user-supplied calibration constants and broaden the range of datasets the extension can process without manual intervention. Finally, a prospective validation study comparing SlicerBoneMat-generated FE model predictions against experimental mechanical testing data, rather than against Bonemat outputs, would provide stronger evidence of clinical utility and support future peer-reviewed publication.

Chapter 6

Conclusions

This thesis presented the design, implementation, and validation of SlicerBoneMat, an open-source extension for 3D Slicer that performs automated CT-based material property mapping to volumetric finite element meshes of bone. The project was motivated by the limitations of existing tools: Bonemat’s closed-source architecture, Windows-only distribution, and absence of an API; MITK-GEM’s lack of published quantitative validation; the commercial licensing costs of Materialise Mimics and Synopsys Simpleware ScanIP; and by the absence of equivalent functionality within the 3D Slicer ecosystem despite documented community demand [KP25, DP23].

The extension implements the full material mapping pipeline from CT image to mapped mesh: phantom calibration coefficient input, the sequential conversion from Hounsfield units through bone mineral density, ash density, and apparent density to Young’s modulus via configurable power-law relationships, and per-element modulus assignment using all three Bonemat algorithm variants with user-selectable Gauss quadrature integration. It supports four volumetric element types, four export formats including a direct FEBio output, in-application visualisation of the mapped material field, a mesh quality statistics panel validated against FEBio Studio, and a named parameter preset system. The extension is implemented in Python within 3D Slicer’s scripted extension architecture, runs natively on Windows, macOS, and Linux, and is distributed as

open-source software under the GNU General Public Licence.

The validation study showed that SlicerBoneMat produces material assignments in close agreement with Bonemat on both test specimens. On the proximal femur at $numSteps = 3$, the RMSE against Bonemat was 0.4212 MPa (less than 0.01% of the mean assigned modulus) with a Pearson correlation coefficient and regression slope both effectively equal to 1.0. On the full femur at $numSteps = 3$, the RMSE was 3.5423 MPa, representing less than 0.08% of the mean assigned modulus, with equally strong correlation. Both results meet the equivalence criteria used in prior comparative validation studies of material mapping implementations. The binning implementation was confirmed to be functionally identical to Bonemat within practical tolerance, and mesh quality statistics matched FEBio Studio’s output at the available comparison precision.

Computational performance scaled as expected – linearly with element count and cubically with integration step count, with $R^2 = 1.0000$ in both cases. The sensitivity analysis showed that the choice of integration step count has a larger influence on assigned modulus values than the choice between SlicerBoneMat and Bonemat, with the within-tool RMSE between $numSteps = 3$ and $numSteps = 4$ being approximately 152 times larger than the tool-to-tool RMSE at matched settings. This finding, to the author’s knowledge, has not previously been quantified in the published literature for this class of material mapping software, and supports consistent reporting of integration parameters alongside any material mapping results, regardless of which tool is used.

Taken together, these results show that SlicerBoneMat meets its design objectives: it replicates Bonemat’s material mapping behaviour, extends its capabilities in platform support, openness, and workflow integration, and (through its direct FEBio export) provides a free, open-source, cross-platform pipeline from CT scan to biomechanical simulation that was previously difficult to achieve without either Bonemat’s platform restrictions or a commercial solver licence. The extension is available on the 3D Slicer Extensions Manager and available to the full Slicer user community.

Future work could prioritise validation on additional anatomical sites and element

types, implementation of cortical-specific mapping to align more closely with Bone-mat V4, and a prospective validation against experimental mechanical testing data, for which the HFValid dataset, with fracture outcomes available for 101 femur specimens, provides a suitable and accessible resource.

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